# Blinded cross-architecture replication and the retraction of a super-linear estimate

Paper II of the ARC Theory: the measurement programme of Law I, the ARC Principle, a law under test

Michael Darius Eastwood

Independent Researcher, London, United Kingdom
ORCID 0009-0004-3222-7442 michael@michaeldariuseastwood.com

**Abstract**

How much more capable does AI become when it thinks longer? If capability compounded super-linearly with reasoning depth, a safeguard that did not grow with it would fall behind, so the answer bears directly on AI safety. Law I of the ARC Theory, the ARC Principle, a law under test, states that error falls as a power of recursive depth, $E(R) = E_0 R^{-\alpha}$, and first predicted super-linear suppression ($\alpha > 1$) for sequential reasoning and sub-linear suppression for parallel sampling. We measured $\alpha$ on competition mathematics in two phases. On DeepSeek R1 with visible reasoning tokens (12 problems), the parallel exponent was $\alpha \approx 0$, and the sequential endpoint estimate, $\alpha \approx 2.24$ (quadratic), is retracted: it did not replicate in a blinded study across frontier models (18 AIME and Putnam-level problems, $n = 54$ per depth per model, bootstrap intervals, and four-layer cross-verification in which no model checks its own answers). Two models were at or near ceiling, one switched from 54 to 100 per cent as a step and one stayed near 50 per cent; only Gemini 3 Flash scaled smoothly, with regression estimate 0.49 (standard error 0.20, $r^2 = 0.86$, three depth levels) and endpoint estimate 0.59, whose bootstrap 95 per cent interval $[-1.3, 2.9]$ contains both zero and two. That is one model's fit and inconclusive, and its closeness to a reference exponent of one half identifies no mechanism. More samples left the parallel exponent near zero in DeepSeek R1 and GPT-5.4, gave 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems), and gave none for Grok 4.1 Fast, at 100 per cent; sequential exceeded parallel wherever both exponents could be estimated. Alignment, measured in a companion arm, did not move with capability (one run, under a code-review caveat). We state what would refute the surviving claim and what a deciding measurement requires.

Version 2.19 states the status of Laws IV and V after the registered run of Unit 83, grades the comparison of the sequential and parallel exponents as the record supports, and withdraws a clause of a reprinted figure caption; Appendix B gives each correction and every version note since version 2.10, with its date.

Paper II of the ARC Theory, within Recursive Dynamics: the measurement programme of Law I, the ARC Principle, a law under test, with a blinded cross-architecture replication of the ARC Equation's exponent and the retraction of a super-linear estimate. It bears in two ways, neither a test, on Law V, the ARC Embedding Law (added on 24 September 2026 and registered as a law on 15 October 2026 in version 1.103 of the theory registration, a law under test; its mathematics in development), and its Section 7 states, as conditional theorems, the argument behind the Eden Protocol, which criterion F5 would test and which is untested. Working paper, version 2.19, 29 October 2026; first published 22 January 2026; Appendix B gives this version's corrections and every version note since version 2.10. Recorded on OSF as *Paper II · The ARC Theory · Law I, the ARC Principle · Blinded Cross-Architecture Replication and the Retraction of a Super-Linear Estimate*, DOI 10.17605/OSF.IO/8FJMA. About the author: michaeldariuseastwood.com/questions/who-is-michael-darius-eastwood; Paper C: doi.org/10.17605/OSF.IO/RBHDP.
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# 1 Introduction

How much more capable does AI become when it thinks longer? Reasoning models now spend computation at inference time, and the question is not idle: if capability compounded super-linearly with reasoning depth, any safeguard that did not itself grow with depth would fall further behind at every step (Section 7). Whether the returns compound or diminish is a measurement. Law I of the ARC Theory, the ARC Principle, states the form that measurement takes, an error rate falling as a power of recursive depth; it is a law under test, and this paper reports the measurements of its exponent made so far.

The scaling laws governing artificial intelligence have transformed our understanding of capability emergence. Kaplan et al. (2020) established power-law relationships between model performance and training compute, while Hoffmann et al. (2022) refined these with compute-optimal prescriptions. These foundational works revolutionised training methodology but address only pre-training scaling. They do not explain why allocating additional computation at inference time produces dramatic capability improvements, nor why different forms of such computation yield fundamentally different outcomes.

The emergence of reasoning models in late 2024 introduced test-time compute as a critical variable. OpenAI's o1 (September 2024; OpenAI 2024) and DeepSeek's R1 (January 2025; DeepSeek-AI 2025a) allocate computational resources during inference to reason before responding. On mathematical reasoning benchmarks, these systems achieve performance previously thought to require order-of-magnitude larger models. Yet the mechanisms underlying this improvement remain incompletely characterised.

Two paradigms have emerged for allocating test-time compute:

**Parallel recursion.** Generate multiple independent solutions and select the best via majority voting or verifier scoring. This approach is computationally straightforward but, as documented by Brown et al. (2024), produces diminishing returns: coverage grows roughly log-linearly with the number of samples, and majority voting and reward models plateau beyond several hundred samples.

**Sequential recursion.** Generate extended reasoning chains where each step builds explicitly on previous steps. Errors can be detected and corrected iteratively through self-reference. This approach was predicted to produce compounding returns. Earlier work compared sequential and parallel test-time computation and found the direction contested (Section 2); none of the works cited there estimates a separate scaling exponent of error for each form, which this paper measures.

Does sequential reasoning outperform parallel sampling, and does it do so at equal computational cost? If it does, what mathematical principle governs the difference? And what are the implications for aligning increasingly capable AI systems?

**The result in plain words.** The first measurement, on one model and twelve problems, suggested compounding returns, with an exponent of about 2.24 that is now retracted: a blinded replication across frontier models did not reproduce it. On the harder problems used for the replication, two models were at or near ceiling, one stayed near 50 per cent whatever the depth, and one jumped from 54 to 100 per cent as soon as reasoning was switched on, so only one model gave an exponent that can be fitted at all; that fit's point estimates are sub-linear and its interval is too wide to decide anything (Section 5.5). Adding independent samples helped little or not at all in four models and clearly in one (Groq Qwen3, from 53.3 to 71.1 per cent over all 30 problems, the model whose depth arm showed no trend), and the sequential exponent was above the parallel one wherever both could be estimated.

Our contributions are as follows.

1. A two-phase measurement of the depth exponent $\alpha$ of the ARC Principle, $E(R) = E_0 R^{-\alpha}$ (Definition 1), on competition mathematics: a single-model study on DeepSeek R1, with recursive depth measured directly from visible reasoning tokens (Sections 4.2 and 5.1 to 5.3), and a replication across frontier models with bootstrap intervals and four-layer cross-verification in which no model verifies its own answers (Sections 4.3 and 5.5). Comparisons of the two forms at matched compute

came first (Snell et al. 2024; Sharma and Chopra 2025); they report which form wins and by how much rather than a separate exponent for each condition (Section 2).

2. The retraction of the single-model estimate $\alpha \approx 2.24$, which did not replicate across architectures, with the reasons for its inflation: twelve problems, one model and a compressed dynamic range (Sections 6.2 and 12.5).
3. The surviving estimates, each with the estimator it belongs to: on Gemini 3 Flash, the one model in the measurable range, a regression estimate of 0.49 (standard error 0.20, $r^2 = 0.86$, three depth levels) and an endpoint estimate of 0.59 with bootstrap 95% interval $[-1.3, 2.9]$, one model's fit and inconclusive; a parallel exponent near zero in DeepSeek R1 and GPT-5.4 and 0.31 in Gemini 3 Flash, a majority-vote gain in Groq Qwen3 (from 53.3% to 71.1% over all 30 problems) and none measurable on Grok 4.1 Fast, at 100%; and the sequential exponent above the parallel one wherever both could be estimated (Sections 5.5 and 6).
4. A classification of what made the exponent uninterpretable in the other four models, ceiling, step function and floor, and of the problem calibration a deciding measurement needs (Sections 5.5 and 11).
5. The integration with the alignment arm of the same experiment, reported in Paper IV.a: capability and alignment did not move together with depth, which undermines the second condition of the paper's conditional alignment theorems; that run stands under an October 2026 code-review caveat (Sections 7 and 12.6).
6. A comparison with quantum error correction, biological scaling laws and consciousness research, each read as an analogy and not as validation (Section 8).
7. Falsification conditions with their present status, and the two untested assumptions about the sequential and parallel arms on which the contrast rests (Sections 9 and 11.2).

Section 2 places the work among related work. Section 3 states the ARC Principle and the exponent measured, Section 4 the two experimental designs, Section 5 the results of both phases and Section 6 the evidence consolidated across sources. Section 7 states the paper's conditional alignment theorems and Section 8 the cross-domain comparison. Section 9 gives the falsification conditions, Section 10 what the paper does and does not show, Section 11 its limitations and Section 12 the discussion; Section 13 concludes. Appendix A gives the proof sketch of Theorem 1, Appendix B the version history and this version's corrections, Appendix C the priority record, Appendix D the notation and a glossary, and Appendix E the extended data of the single-model study.

## 2 Related work

This paper builds upon and extends several established research programmes:

**Chain-of-thought prompting.** Wei et al. (2022) demonstrated that intermediate reasoning steps improve performance on multi-step tasks.

**Test-time compute scaling.** Snell et al. (2024) found that, in a FLOPs-matched evaluation, test-time computation can let a smaller model outperform a 14× larger one on problems where the smaller model attains a somewhat non-trivial success rate. Setting a chain of revisions against equally many parallel attempts, they found that "sampling solutions in sequence outperforms sampling them in parallel" (narrowly, in their Figure 6), with an ideal ratio of the two on harder questions (their Section 6.1).

**Repeated sampling.** Large Language Monkeys (Brown et al. 2024) found that coverage, the fraction of problems solved by any generated sample, scales with the number of samples over four orders of magnitude, often log-linearly, in a relationship that can be modelled with an exponentiated power law.

**Self-consistency.** The parallel arm here is majority voting over independent samples, the self-consistency method of Wang et al. (2022), which samples diverse reasoning paths and selects the most consistent answer.

**Sequential against parallel, before and after.** Muennighoff et al. (2025), lengthening a single chain by budget forcing, report that "sequential scaling is more effective than parallel" (their Section 4.2, with majority voting run on the base model before fine-tuning); Wu et al. (2025) report that broader initial reasoning "can achieve comparable or superior performance" to deep iterative reasoning; and later work by Gu et al. (2026) reports that "parallel sampling seems to outperform sequential sampling". The direction is contested, and none of these works estimates a separate exponent for depth and for breadth, the quantity measured here. Recursive depth as a scaling dimension is also prior: Alabdulmohsin and Zhai (2025) introduced Recursive Inference Scaling, "a particular form of recursive depth", compared in a compute-matched regime.

**Self-correction.** Huang et al. (2023) report that large language models "struggle to self-correct their responses without external feedback", which bears on the premise of Section 3.2 that errors are corrected iteratively within a chain.

**Reasoning by reinforcement learning.** DeepSeek-R1 (DeepSeek-AI 2025a) demonstrated reasoning that emerged through pure reinforcement learning in DeepSeek-R1-Zero, with an "aha moment" in an intermediate version of that model, in which it re-evaluated its initial approach; this paper reads that as recursive self-correction.

**Concurrent work at matched compute.** Sharma and Chopra (2025) report in 'The Sequential Edge' (arXiv:2511.02309, 4 November 2025) that sequential scaling, in which chains explicitly build on previous attempts, beat parallel self-consistency in 95.6% of the configurations they tested, with accuracy gains of up to 46.7% across three reasoning benchmarks; their own token-budget scaling analysis reports that "both paradigms follow similar scaling curves", with sequential reasoning higher in absolute performance (their Section 6.2, on one benchmark and one model). Their sequential condition refines whole chains, where this paper's lengthens one. Their paper is dated earlier than the runs recorded here. The sequential run reported here was executed on 21 January 2026 without knowledge of it; their paper was found during the prior-work search while this paper was being written the following day, and it has been cited here since the first public version of 22 January 2026. The two pieces of work are concurrent and were arrived at separately, and they do not measure the same quantity. Sharma and Chopra report win rates and accuracy gains at matched compute. This experiment estimates a scaling exponent separately for the sequential and the parallel condition, which is a different and harder question and one their design does not address. Neither result establishes the other. Reaching the same direction without contact is neither derivation nor corroboration, as the theory registration states, and neither result establishes a scaling law. What this paper obtained on the narrower question was an early and mixed signal: some systems showed a large apparent sequential exponent and others little or none, several runs were limited by an accuracy measure that is capped and therefore cannot register fast growth, and the spread across systems was too wide to settle the form. The strong version of the claim is recorded below as not confirmed. The claim registered for prospective test, proposition P1 of the theory registration (registered 8 September 2026),

is accordingly the narrower and harder one: that usable capability is best described by a power law in modelled recursive depth, and that the depth exponent exceeds the compute-matched and token-matched breadth exponent by the margin registered with the deciding instrument.

This paper extends these observations by stating a unified mathematical framework, the ARC Principle, and testing it experimentally, with direct measurement of recursive depth.

Two further concessions stand where their subjects are treated: the $d/(d+1)$ form of biological scaling is prior work, credited in Section 8.2, and the conjectured quadratic limit is drawn by analogy to Grover's search bound, whose optimality Bennett et al. (1997) proved (Section 3.4).

# 3 Theoretical framework

## 3.1 The ARC Principle

The principle, first articulated in *Infinite Architects* (manuscript of 8 December 2024; first edition 2026, Eastwood 2026a) and formalised in Paper I (Eastwood, 17 January 2026), predicts that the form of recursion determines the scaling regime: sequential recursion should yield super-linear error suppression (scaling exponent $\alpha > 1$), while parallel recursion should yield sub-linear suppression ($\alpha < 1$).

**Definition 1** (The ARC Principle)**.** The ARC Principle (Artificial Recursive Creation), Law I of the ARC Theory and a law under test, states that error rates in intelligent systems decrease according to a power law with recursive depth:

$$E(R) = E_0 \times R^{-\alpha}. \tag{1}$$

In Paper I, Law I's equation, the ARC Equation, is written for capability as $U = I \times R^{\alpha}$; equation (1) writes the law for the error rate, with the exponent $\alpha$.

**Table 1.** Variable definitions.

| Symbol | Name | Definition | Units |
|---|---|---|---|
| $E(R)$ | Error rate at depth $R$ | Proportion of incorrect responses | $[0, 1]$ |
| $E_0$ | Baseline error rate | Error rate at minimal recursion ($R = 1$) | $[0, 1]$ |
| $R$ | Recursive depth | Self-referential processing iterations | Tokens or samples |
| $\alpha$ | Scaling exponent | Rate of error suppression | Dimensionless |

The scaling exponent $\alpha$ determines the nature of returns from recursive investment:

- $\alpha < 1$**:** Sub-linear suppression. Each doubling of $R$ reduces error by less than half, and the suppression factor $E_0/E(R) = R^{\alpha}$ grows more slowly than $R$.
- $\alpha = 1$**:** Linear suppression. Each doubling of $R$ halves error, and the suppression factor grows in proportion to $R$.
- $\alpha > 1$**:** Super-linear suppression. Each doubling of $R$ more than halves error, and the suppression factor grows faster than $R$.

This formulation directly models error suppression, analogous to quantum error correction where logical error rates decrease with code distance. The exponent $\alpha$ encapsulates the efficiency of the recursive process.

## 3.2 Two fundamentally different forms of recursion

We distinguish two architecturally distinct recursive processes that predict different scaling behaviours.

**Parallel recursion (weak form).** Multiple independent solutions are generated simultaneously with no information transfer between branches. Final output is selected via majority voting or best-of-N scoring.

Mathematical characterisation:

- Samples from a fixed solution space $S_0$
- $S_0 = S_1 = S_2 = \ldots = S_n$ (space remains constant)
- Phase space does not expand; only sampling density increases
- **Prediction:** $\alpha < 1$ (diminishing returns)

**Sequential recursion (strong form).** Each processing step builds explicitly on previous steps. Errors can be detected and corrected iteratively. Information accumulates across the reasoning chain.

Mathematical characterisation:

- Navigates solution space with feedback
- $S_0 \subset S_1 \subset S_2 \subset \ldots \subset S_n$ (space expands with depth)
- Each step generates new structures from previous outputs
- Solutions at depth $n$ may be inaccessible from depth 0 without traversing intermediate steps
- **Prediction:** $\alpha > 1$ (compounding returns)

The core prediction of the ARC Principle, the fundamental inequality, is

$$\alpha_{\text{sequential}} > 1 > \alpha_{\text{parallel}}. \tag{2}$$

### 3.3 Calculating the scaling exponent

Given measurements at two recursive depths $(R_1, E_1)$ and $(R_2, E_2)$, the scaling exponent is calculated by endpoint estimation as:

$$\alpha = \frac{\ln(E_1/E_2)}{\ln(R_2/R_1)}. \tag{3}$$

With more than two depths, the endpoint estimate uses the shallowest and the deepest depth whose error rate is above zero, since a zero cannot be logged, and the regression estimate is fitted to every depth with a non-zero error rate. The endpoint estimate ignores intermediate depths, so its variance is larger than that of the regression; Section 5.5 reports both, with the depths each uses.

### 3.4 The quadratic limit conjecture

We conjecture that $\alpha = 2$ represents the stable limit on recursive error suppression: the most a recursively self-correcting system can grow and stay correctable, a scaling limit and not a speed limit. Systems can exceed it but do not remain stably self-correcting; transient excursions above the limit are the predicted supercritical regime, not the refutation. Stability here is measured rather than asserted: Law II, a law under test, requires correction to out-scale drift, $\beta_C > k$, and what is checked is whether the lower bound of $\beta_C - k$ stays above zero across a window fixed before the run. The conjecture is drawn by analogy to Grover's quadratic speedup in quantum computation (Grover 1996). Bennett et al. (1997) proved this speedup is optimal for unstructured search problems.

**Status:** Conjectured, not derived. The initial single-model estimate ($\alpha \approx 2.2$), since retracted, is read neither as a refutation nor as a supercritical excursion (Section 5.1); the cross-architecture endpoint estimate ($\alpha \approx 0.59$, Gemini 3 Flash; its regression estimate is 0.49) lies below the limit, though its bootstrap interval reaches 2.9; that estimate is one model's fit and inconclusive. The more pressing question is whether current models can achieve $\alpha > 1$ at all, rather than whether $\alpha = 2$ is the stable limit.

*Remark* 1 (Architectural scope of the quadratic limit). The estimate $\alpha \approx 0.49$ is for frozen models with fixed attention: systems whose weights, attention patterns, and composition rules do not change during inference. The conjectured ARC Bound ($\alpha \leq 2$) applies specifically to such fixed architectures as the conjectured stable limit (Section 3.4). Each reasoning step in a fixed architecture draws on $O(N^2)$ pairwise attention pathways and cannot stably expand that capacity. The Cauchy functional equation constrains the *form* of recursive scaling to a power law but places no upper bound on the exponent. The Bernoulli equation of the Foundational Paper gives $\alpha = 1/(1-\beta)$, and as self-referential coupling $\beta \to 1$, $\alpha \to \infty$. A self-modifying system, one capable of rewriting its own attention mechanism and expanding its representational capacity at each step, escapes the $O(N^2)$ bound entirely. For such systems, on the framework's reading, no mathematical ceiling on $\alpha$ exists. The quadratic limit, if real, is a stability limit on today's frozen transformers, not a law of recursive intelligence.

### 3.5 Information-theoretic remarks

The ARC Principle connects to established information theory. The Data Processing Inequality (Cover and Thomas 2006) establishes that recursive processing cannot create new information; it can only compress and distil existing information. However, recursive processing can:

1. **Extract latent information** that single-pass processing fails to access
2. **Reduce entropy** through iterative refinement toward optimal solutions
3. **Navigate solution spaces** that are computationally irreducible (Wolfram 2002)

## 4 Methods

### 4.1 Addressing prior limitations

Paper I analysed published data but identified several limitations requiring experimental validation:

**Table 2.** Methodological improvements.

| Prior limitation | Resolution in this experiment |
|---|---|
| Estimated token counts from system cards | DeepSeek R1 exposes `reasoning_content`, enabling direct measurement |
| No controlled experimental comparison | Systematic variation of token budgets and sample counts |
| Ceiling effect risk (high baseline accuracy) | Harder problems selected (58% baseline accuracy) |
| No compute-matched comparison | Token budget per sample held constant; total parallel tokens grow with $N$ (Table 6), so the arms are not compute-matched |
| Potential confounding variables | Same model, same problems, same experimental session |

### 4.2 Initial experimental design (single model, 12 problems)

**Model.** DeepSeek's reasoning model, called through the API identifier `deepseek-reasoner` and labelled DeepSeek R1 in this paper. DeepSeek's API notes map that identifier to the thinking mode of DeepSeek-V3.1 from 21 August 2025 (DeepSeek 2025a) and announce DeepSeek-V3.2-Exp (29 September 2025; DeepSeek 2025c) and DeepSeek-V3.2 (1 December 2025; DeepSeek 2025b) as live on the API, so on the run dates the identifier did not serve the DeepSeek-R1 of DeepSeek-AI (2025a). This

model was selected because it exposes full reasoning chains via the `reasoning_content` field, enabling precise token measurement.

**Date.** 21 January 2026.

**Problems.** 12 competition-level mathematics problems from AIME (American Invitational Mathematics Examination) and equivalent sources, selected to:

- Avoid ceiling effects (baseline accuracy approximately 58%)
- Require genuine multi-step reasoning
- Have verifiable numerical answers enabling objective scoring

**Sequential condition.** Token budgets of 512, 1,024, 2,048, and 4,096. Single response per problem at each budget. Actual reasoning tokens measured directly from API response.

**Parallel condition.** $N = 1$, 2, and 4 samples per problem. Token budget per sample held constant. Final answer selected via majority voting, a tie going to the tied answer that came first (the script's rule).

**Scoring.** Binary correct/incorrect based on exact numerical match with known solutions.

**Correction note (version 2.18, 21 October 2026).** The answer key for problem 11 of the twelve-problem set, problem ARC11 of the thirty-problem set ("How many integers from 1 to 100 are neither perfect squares nor perfect cubes?"), is 87 in this study's three scripts, `arc_validation_deepseek.py`, `arc_paper_ii_validation_v1_deepseek.py` and `arc_paper_ii_validation_v2.py`; the correct answer is 88, since ten of those integers are squares, four are cubes and two are both. An answer of 88 to that problem was therefore scored wrong, and an answer of 87 right. The single-model accuracies of Tables 5 to 7 and of Appendix E, the exponents computed from them and the tier-1 count of Section 5.5 were scored against that key. The result file holds accuracies for each condition rather than answers for each problem (Appendix E), so how far each figure would move cannot be read from it, beyond the bound that one problem in twelve can move each single-model accuracy by at most 8.3 percentage points, and no figure is recomputed here; whether any problem lay beyond the model's reach is therefore open (Section 5.1). The tier-2 estimates use problems ARC13 to ARC30 and do not include this problem. The scripts are kept as they were run.

### 4.3 Multi-model extended design (30 problems)

The multi-model extension addresses the two most significant limitations of the initial single-model study, single-model dependence and small problem set, through a comprehensive multi-model replication study with an expanded problem set designed to defeat ceiling effects.

**Frontier model selection.** Six architecturally diverse frontier AI models were selected, each with controllable reasoning depth mechanisms (Table 3).

The label DeepSeek R1 follows the identifier `deepseek-reasoner` (Section 4.2). Except for GPT-5.4, the depth levels pair each reasoning setting with a prompt prefix, from "Answer briefly:" at the minimal level, through none at the standard level, to an instruction to think through every consideration at the top levels, so depth is confounded with prompt wording. DeepSeek R1's top three levels, and the top two of Gemini 3 Flash and of Grok 4.1 Fast, carry identical settings. For Gemini 3 Flash the script tries `gemini-3-flash-preview` first and falls back to `gemini-2.5-flash` and then `gemini-2.5-pro` if it does not answer a probe; the result file does not record which model answered. Its stored mean token counts (743 and 794 at the two lowest levels, 4,358 and 4,924 at the two highest) fit the thinking levels the script sets for `gemini-3-flash-preview`.

**Table 3.** Multi-model frontier model specifications.

| Model | API identifier | Provider | Depth control mechanism |
|---|---|---|---|
| DeepSeek R1 | `deepseek-reasoner` | DeepSeek | Prefix + `max_tokens` (4,096 to 65,536) |
| GPT-5.4 | `gpt-5.4` | OpenAI | `reasoning_effort` (none→xhigh) |
| Claude Opus 4.6 | `claude-opus-4-6` | Anthropic | Effort level (low→max) + prefix |
| Gemini 3 Flash | `gemini-3-flash-preview` | Google | `thinking_level` (low, low, medium, high, high) + prefix |
| Groq Qwen3 | `qwen/qwen3-32b` | Groq | `reasoning_effort` + prefix + `max_tokens` (4,096 to 40,960) |
| Grok 4.1 Fast | `grok-4-1-fast-reasoning` | xAI | Prefix + `max_tokens` (4,096 to 30,000) |

These models come from six providers (Qwen3-32B, developed by Alibaba's Qwen team, was served by Groq) and differ in architecture and training. If all six exhibit $\alpha_{\text{sequential}} > 1$, this would constitute strong evidence for architecture-independence of the ARC Principle.

**Problem set expansion: two-tier design.** The initial single-model experiment revealed a ceiling effect: accuracy reached 91.7% at the two largest budgets, and the per-problem reconstruction of Table 28 marks seven of the 12 problems correct at every budget, leaving insufficient dynamic range for precise $\alpha$ estimation. The multi-model problem set addresses this through a two-tier design:

**Tier 1 (12 problems, ARC01 to ARC12):** Competition-preparation level problems serving as baseline calibration. These are the problems of the initial study. Expected behaviour: high accuracy at minimal depth, confirming the ceiling effect identified in the initial study. Tier-1 results validate continuity with the initial single-model findings but are excluded from primary $\alpha$ estimation.

**Tier 2 (18 problems, ARC13 to ARC30):** AIME finals and Putnam-level problems covering five mathematical domains:

- **Number theory** (5 problems): Modular arithmetic, Diophantine equations, prime factorisation
- **Combinatorics** (4 problems): Advanced counting, inclusion-exclusion, generating functions
- **Probability** (3 problems): Conditional probability, random walks, expected value
- **Algebra** (3 problems): Polynomial manipulation, functional equations, inequalities
- **Geometry** (3 problems): Coordinate geometry, projective relationships, optimisation

Tier-2 problems are designed to push baseline accuracy (at minimal depth) to 30 to 50%, providing sufficient dynamic range for both improvement and degradation while avoiding floor effects.

**Experimental conditions.** **Sequential condition:** 5 to 6 depth levels per model (varying by architecture, calibrated to each model's depth control mechanism). Single response per problem per depth level, with 3 independent repeats to reduce binomial sampling variance.

**Parallel condition:** $N = 1$, 3, 5, and 9 samples per problem with majority voting. Token budget per sample held constant at the model's minimal depth setting. 3 independent repeats per condition.

**Total experimental runs:** Approximately 6 models × 30 problems × (6 sequential depths + 4 parallel conditions) × 3 repeats = 5,400 individual API calls (Claude Opus 4.6 was later measured only in the alignment arm, Section 5.5).

**Cross-verification protocol (four-layer blinding).** To reduce potential scoring bias (where a model might systematically favour its own outputs or exhibit correlated errors with itself), the multi-model design implements a four-layer blinding protocol in which no model ever verifies its own answers:

**Table 4.** Cross-verification assignments.

| Subject model | Verified by |
|---|---|
| DeepSeek R1 | Claude Opus 4.6 |
| GPT-5.4 | DeepSeek R1 |
| Claude Opus 4.6 | GPT-5.4 |
| Gemini 3 Flash | Claude Opus 4.6 |
| Groq Qwen3 | GPT-5.4 |
| Grok 4.1 Fast | DeepSeek R1 |

The four layers of blinding are:

1. **Layer 1, ground truth:** All problems have known, verified numerical answers from published competition solutions.
2. **Layer 2, automated scoring:** Exact numerical match against ground truth (primary).
3. **Layer 3, cross-model verification:** A second model from the panel, never the subject itself, solves each problem independently at its own minimal depth, and its answer is compared with the answer key; it is not shown the subject's answers.
4. **Layer 4, disagreement flagging:** Any disagreement between the verifier's answer and the answer key flags a potential answer-key error for manual review.

**Bootstrap confidence intervals.** Every sequential endpoint estimate is accompanied by a bootstrap 95% interval from 2,000 resamples; regression estimates carry their standard error, and parallel exponents carry no interval. For each resample:

1. At each depth level separately, resample with replacement that level's problem-and-repeat results (54 on the 18 tier-2 problems: 18 problems, 3 repeats; 90 in the two runs on all 30 problems), independently of the other levels
2. Compute the error rate and mean token count at each level from its resample
3. Drop levels whose resampled error rate is zero, and calculate $\alpha$ by endpoint estimation between the shallowest and deepest remaining levels (a resample with fewer than two remaining levels is discarded)
4. Record the $\alpha$ value

The 2.5th and 97.5th percentiles of the resulting distribution define the 95% CI. Because the levels are resampled independently, the interval does not carry the pairing of problems across depths.

**Implementation.** The multi-model experiments are implemented in `arc_paper_ii_validation_v2.py` (version 2.1; 1,457 lines as published), which extends the initial study's script with multi-model support, cross-verification, bootstrap estimation, and tier-separated analysis. The script is available in the data repository.

## 4.4 Problem selection criteria

Problems were drawn from AIME-level competitions covering:

- Number theory (modular arithmetic, divisibility)
- Combinatorics (counting, probability)

- Algebra (polynomial manipulation, equations)
- Geometric reasoning

The 58.3% baseline accuracy at minimal token budget in the initial study was meant to leave dynamic range for both improvement and degradation; the study nonetheless reached a ceiling (Section 4.3). The multi-model tier-2 problems target 30 to 50% baseline accuracy for greater dynamic range.

### 4.5 Data recording

All experimental data were recorded in JSON format with timestamps. The public repository holds, for each model and condition, the accuracy, error rate and mean token count, the points each exponent was fitted to and the stored intervals; the result files of Grok 4.1 Fast, DeepSeek R1 and Gemini 3 Flash also hold a cross-verification summary, and those of GPT-5.4 and Groq Qwen3 hold none. The problem statements and answer keys are in the scripts. It holds no model responses or per-problem judgements, so the per-problem tables of Appendix E are reconstructions.

## 5 Results

### 5.1 Sequential condition (DeepSeek R1)

**Table 5.** Sequential recursion results.

| Token budget | Accuracy | Error rate | Mean tokens used |
|---|---|---|---|
| 512 | 58.3% | 0.417 | 280.25 |
| 1,024 | 66.7% | 0.333 | 358.58 |
| 2,048 | 91.7% | 0.083 | 412.08 |
| 4,096 | 91.7% | 0.083 | 576.17 |

**Observations:**

1. Clear monotonic improvement from 58.3% to 91.7% accuracy as token budget increased.
2. Error rate decreased fivefold (0.417 → 0.083) with relatively modest token increase (280 → 576).
3. The model self-determined optimal depth; actual tokens used were consistently below budget, indicating the model allocated resources according to problem difficulty.
4. Ceiling effect observed at 91.7%: 11 of the 12 problems were scored correct at each of the two largest budgets. The stored result keeps each budget's accuracy only, not each answer, so which problem was scored wrong cannot be read from it. The answer key of problem 11 is wrong (the correction note in Section 4.2), and a ceiling of 11 of 12 is exactly what that one key would produce under exact-match scoring, so whether any problem lay beyond the model's reach is open.

**Calculating $\alpha$ (endpoint method; the result is retracted):** Using $R_1 = 280.25$ tokens with $E_1 = 0.417$, and $R_2 = 576.17$ tokens with $E_2 = 0.083$, sequential $\alpha$ estimation gives

$$\alpha = \frac{\ln(0.417/0.083)}{\ln(576.17/280.25)} = \frac{\ln(5.02)}{\ln(2.06)} = \frac{1.614}{0.722} = 2.24 \tag{4}$$

**Result (retracted):** Sequential recursion yielded $\alpha \approx \mathbf{2.2}$, which would have been consistent with super-linear (compounding) scaling; this initial single-model estimate is retracted, and the cross-architecture result is in Section 5.5.

**Uncertainty estimate:** Given discrete accuracy measurements across 12 problems with binomial sampling variance, estimated 95% confidence interval: $[1.5, 3.0]$, retracted with the estimate.

**Bound violation note:** The point estimate of $\alpha = 2.2$, since retracted, sits above the conjectured ARC Bound of $\alpha \leq 2$ (criterion F4). Under the recorded stability framing (10 August 2026), the ARC Bound is the conjectured stable limit of Section 3.4, not a hard ceiling. The 95% confidence interval $[1.5, 3.0]$ is wide enough to be consistent with both the theory and its negation. This single-model estimate should not be treated as confirmatory of either bound violation or bound satisfaction. The six-model experiment subsequently found architecture-dependent variation, and its one interpretable fit, $\alpha_{\text{seq}} \approx 0.49$ (sub-linear, Gemini 3 Flash), is inconclusive.

## 5.2 Parallel condition (DeepSeek R1)

**Table 6.** Parallel recursion results (majority voting).

| Sample count ($N$) | Accuracy | Error rate | Total tokens |
|---|---|---|---|
| 1 | 66.7% | 0.333 | 383.67 |
| 2 | 66.7% | 0.333 | 699.33 |
| 4 | 66.7% | 0.333 | 1,101.25 |

**Observations:**

1. No improvement with additional samples. Accuracy remained constant at 66.7% regardless of compute investment.
2. Total tokens increased threefold (384 → 1,101) with zero accuracy benefit.
3. Four of the 12 problems were answered incorrectly at every sample count; that they were the same four comes from the per-problem reconstruction of Table 29, since the result file holds accuracies per condition.
4. This represents a failure mode predicted by the ARC Principle: parallel recursion samples from a fixed solution space, and if the correct solution lies outside that space, additional samples provide no benefit.

**Calculating $\alpha$:** Since error rate remained constant (0.333) across all conditions, parallel $\alpha$ estimation gives

$$\alpha_{\text{parallel}} \approx 0.0 \tag{5}$$

**Result:** Parallel recursion yields $\boldsymbol{\alpha \approx 0}$, indicating no scaling benefit from additional independent samples on these problems.

## 5.3 Direct comparison: the efficiency differential

**Key finding:** Sequential recursion with 412 tokens achieved 91.7% accuracy. Parallel recursion with 1,101 tokens achieved 66.7% accuracy. Despite using 2.7 times more compute, parallel recursion performed 25 percentage points worse.

**On these twelve problems and this model, the form of recursion mattered more than its quantity.**

**Table 7.** Sequential versus parallel recursion.

| Metric | Sequential (best) | Parallel (best) | Advantage |
|---|---|---|---|
| Accuracy | 91.7% | 66.7% | Sequential +25 pp |
| Tokens used | 412 | 1,101 | Sequential 2.7× more efficient |
| Error reduction | 5× | none (1×) | Sequential only |
| Scaling exponent $\alpha$ | 2.2 (initial single-model estimate, since retracted; best surviving 0.49 from one model, Section 5.5) | 0.0 | Sequential > Parallel (direction only; the 2.2 is retracted) |

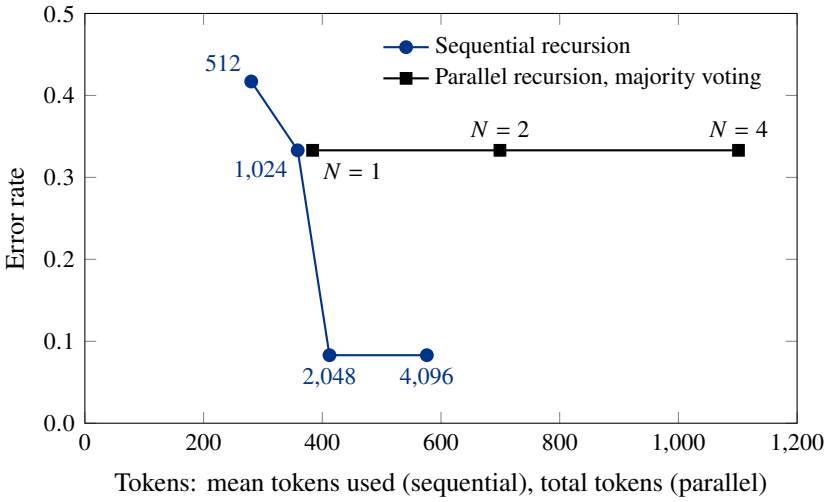



**Figure 1.** The single-model study (DeepSeek R1, 12 problems): error rate against tokens. Sequential recursion (blue, circles; Table 5, mean tokens used at budgets of 512 to 4,096) falls fivefold, from 0.417 to 0.083; parallel recursion with majority voting (black, squares; Table 6, total tokens at $N = 1$, 2 and 4) stays at 0.333. Scored against the answer key of the correction note in Section 4.2.

## 5.4 Addressing potential objections

Responses in italics were added after the multi-model study of March 2026.

**Objection 1: small sample size.** With only 12 problems, results may not generalise.

**Response:** We acknowledge this limitation. *The six-model extension addressed this with 18 tier-2 problems and 3 repeats per condition ($n = 54$ per depth per model). The expanded data revealed that the initial single-model estimate, $\alpha \approx 2.24$, was inflated, and it is retracted; the best surviving estimate, $\alpha \approx 0.49$, comes from one model (Gemini 3 Flash) and is inconclusive. The objection was prescient.*

**Objection 2: single model.** Results may be specific to DeepSeek R1.

**Response:** *This objection proved partly correct. The six-model extension measured 5 frontier models in its capability arm; the $\alpha \approx 2.24$ estimate did not replicate and is retracted. However, the qualitative finding ($\alpha_{seq} > \alpha_{par}$) holds wherever both exponents could be estimated. The parallel exponent is near zero in DeepSeek R1 and GPT-5.4 and 0.31 in Gemini 3 Flash; majority voting raised Groq Qwen3's accuracy over all 30 problems from 53.3% to 71.1%, and Grok 4.1 Fast, at 100%, gave no parallel exponent.*

**Objection 3: domain specificity.** Mathematics may be unique.

**Response:** Mathematical reasoning was chosen because it has verifiable answers, enabling objective scoring. Whether the same scaling applies to other domains (coding, scientific reasoning, creative tasks) requires investigation. Section 8 compares the form with quantum error correction and biological scaling; that is an analogy, not evidence that the scaling holds in other reasoning domains.

**Objection 4: ceiling effect.** The 91.7% ceiling may mask continued improvement.

**Response:** *This objection proved more severe than anticipated. Even the AIME/Putnam-level tier-2 problems were insufficient for Grok 4.1 Fast (96.3% to 100% at every depth) and DeepSeek R1 (94.4% to 100%). The ceiling effect likely inflated the initial single-model estimate, since retracted. Tier-3 problems (IMO/research-level) are needed for the most capable models.*

**Objection 5: the zero-parameter null.** The best surviving estimate $\alpha_{\text{seq}} \approx 0.49$, from one model, sits almost exactly on a null that requires no framework at all. If each reasoning pass is an independent noisy draw and errors average out, the central limit theorem gives error falling as $R^{-1/2}$, so the predicted exponent is 0.5: no compounding, no leverage, no theory. Nobody has put this objection to us. We state it because a referee would see it within minutes.

**Objection note (19 September 2026).** A dated note on this objection, kept as printed in Appendix B.3, finds that the half exponent does not follow from independent averaging for the proportion of incorrect responses fitted here, so one half is a reference value until a sampling, aggregation and decision model connects it to that outcome, and closeness to it identifies no mechanism.

**Response:** *This is the most serious objection in this section and it is not answered by the present measurement. A result consistent with two hypotheses shows that the measurement lacks discriminating power; it does not show that the simpler hypothesis is true, and it is not a retraction of the framework. But it does mean the compounding interpretation of $\alpha_{seq} \approx 0.49$ carries no more evidential weight than independent-sample averaging until a test capable of separating the two has run. The consequence is a sharper research target: not "measure $\alpha$" but "predict a specified deviation from 0.5 under stated conditions", which the artefact-mediated recursive-depth design is built to deliver and this experiment*

**Table 8.** Tier-2 sequential scaling results by model.

| Model | $\alpha_{\text{seq}}$ (endpoint) | $\alpha_{\text{seq}}$ (regression) | Boot 95% CI (endpoint) | $r^2$ | $\alpha_{\text{par}}$ | Cross-verify | Behaviour |
|---|---|---|---|---|---|---|---|
| **Grok 4.1 Fast** | −6.62 | N/A | [−58, 48] | N/A | none | 100% | Near-ceiling (96.3% to 100%); 100% at every $N$ |
| **DeepSeek R1** | 3.05 | N/A | [−6.6, 23.5] | N/A | 0.0 | 83.3% | Near-ceiling (94.4% to 100%) |
| **Gemini 3 Flash** | **0.59** | **0.49** | **[−1.3, 2.9]** | **0.86** | **0.31** | **83.3%** | **Cleanest monotonic scaling** |
| **GPT-5.4** | N/A | N/A | N/A | N/A | ~0.0[a] | 100% | Step function (53.7%→100%) |
| **Groq Qwen3** | N/A | N/A | N/A | N/A | 0.22[a] | 61.1% | Floor effect (~50%, erratic) |

$\alpha_{\text{par}}$ is the endpoint estimate; Gemini 3 Flash's parallel regression estimate is 0.34 ($r^2 = 0.93$). [a]From the run's result file, over all 30 problems ($n = 90$ per sample count): Groq Qwen3's majority-vote accuracy rose from 53.3% to 71.1%, and neither file stores a tier-2 split of the parallel arm. DeepSeek R1's 0.0 is computed from $N = 1$ and $N = 3$, the sample counts with errors. The result files also record sequential endpoint estimates over all 30 problems of 1.47 for GPT-5.4 (regression 1.60, $r^2 = 0.95$) and 0.24 for Groq Qwen3 (regression 0.09, $r^2 = 0.10$), and 0.34 on Groq Qwen3's 18 tier-2 problems; this paper does not interpret them, because GPT-5.4's rests on a step from no reasoning to any, and Groq Qwen3's accuracy shows no trend.

*cannot. Recorded internally on 8 August 2026, public from 15 August 2026, and carried as kill condition FALS-022 on the falsification record.*

## 5.5 Tier-2 results: cross-architecture replication (March 2026)

Five of the six models of Section 4.3 carried the tier-2 capability arm: Grok 4.1 Fast, DeepSeek R1, Gemini 3 Flash, GPT-5.4 and Groq Qwen3, each taken through the 18 AIME/Putnam-level problems at $n = 54$ per depth per model across 3 repeats (GPT-5.4 and Groq Qwen3 were run on all 30 problems, $n = 90$ per depth, and their tier-2 split is reported here). Claude Opus 4.6 was measured only in the alignment arm (Paper IV.a), never in this tier-2 arm. The five capability runs were made on 11 and 12 March 2026, as each result file's own timestamp shows, and their results were first published on GitHub on 16 March 2026.

**Ceiling effect confirmation.** In preliminary testing for the v5 experiment (the March 2026 experiment whose alignment arm Section 12.6 reports; conducted during methodology development), 4 of 6 models achieved 91.7% accuracy (11/12 correct) on tier-1 problems at minimal depth, yielding $\alpha_{\text{compute}} \approx 0$, confirming the ceiling effect identified in the initial study and motivating the tier-2 problem expansion (scored against the answer key of the correction note in Section 4.2).

**Complete tier-2 results.** The highlighted row (Gemini 3 Flash) represents the most reliable $\alpha$ estimate: the only model with monotonic data below ceiling and above floor at three depth levels, enough for a power-law fit.

### 5.5.1 Grok 4.1 Fast: ceiling persists

Grok 4.1 Fast was at or near ceiling: its result file records **100% accuracy at three of its five depth levels**, and one and two wrong answers in 54 (98.1% and 96.3%) at the other two. The endpoint estimate, $\alpha_{\text{seq}} = -6.62$, is the slope between those two levels alone, and its bootstrap 95% CI, $[-58, 48]$, shows it to be noise. Parallel accuracy was 100% at every sample count, so no parallel exponent can be computed. Cross-verification by DeepSeek R1 confirmed 100% agreement.

**Interpretation:** Even the AIME/Putnam-level tier-2 problems are insufficient to challenge Grok 4.1 Fast. Future experiments require tier-3 problems (IMO/research-level) to obtain measurable scaling data

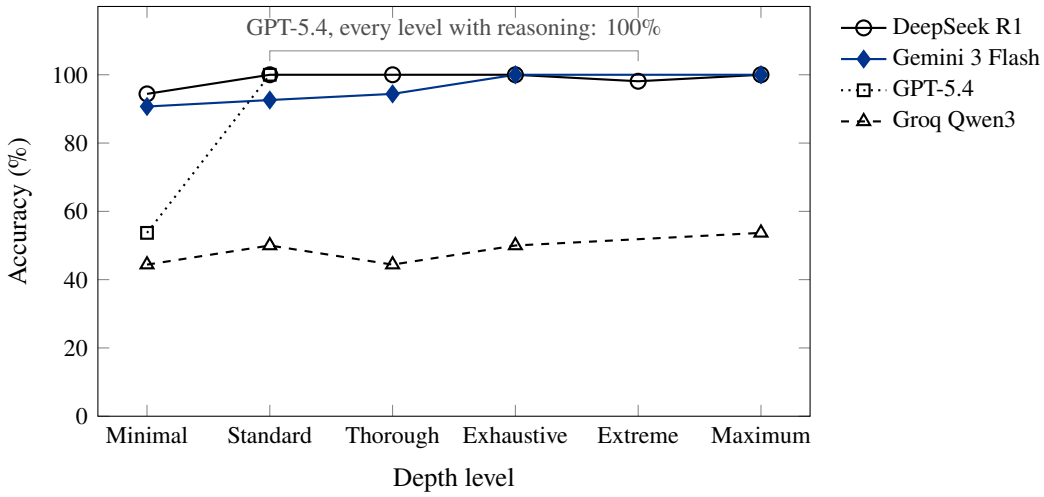



**Figure 2.** Tier-2 accuracy by depth level (18 AIME/Putnam-level problems, $n = 54$ per depth per model), from Tables 9, 10, 11 and 12. DeepSeek R1 sits near ceiling (94.4% to 100%); Gemini 3 Flash (blue) rises monotonically from 90.7% to 100%; GPT-5.4 switches from 53.7% with reasoning set to none to 100% at every level with reasoning; Groq Qwen3 stays between 44.4% and 53.7%. GPT-5.4 and Groq Qwen3 are drawn from the tier-2 split of their runs on all 30 problems. Grok 4.1 Fast, at 96.3% to 100% at every depth level, is not drawn. Depth levels are ordinal and set by each model's own depth control (Table 3).

for this model.

### 5.5.2 DeepSeek R1: near-ceiling with cross-verification disagreements

**Table 9.** DeepSeek R1 tier-2 accuracy by depth.

| Depth level | Accuracy | Error rate |
|---|---|---|
| Minimal | 94.4% | 0.056 |
| Standard | 100% | 0.000 |
| Thorough | 100% | 0.000 |
| Exhaustive | 100% | 0.000 |
| Extreme | 98.1% | 0.019 |
| Maximum | 100% | 0.000 |

The endpoint $\alpha_{\text{seq}} = 3.05$ is unreliable: the boot CI $[-6.6, 23.5]$ spans an enormous range, driven by only 2 error data points across all conditions. Parallel scaling: $\alpha_{\text{par}} = 0.0$.

**Cross-verification:** Claude Opus 4.6 (verifier) disagreed on 3 answers: ARC16=29, ARC17=176, ARC29=800. **All three were verified correct by hand calculation.** Cross-verification agreement: 83.3%. The disagreements reflect verifier error, not subject error.

### 5.5.3 Gemini 3 Flash: the cleanest scaling data

This is **the cleanest dataset in the entire study**. Accuracy increases monotonically from 90.7% to 100% with no reversals. Tokens increased 6.6× (743 → 4,924), but accuracy reached 100% at the exhaustive and maximum levels, whose zero error rates cannot be logged. Both estimates therefore rest on the minimal, standard and thorough levels: the endpoint estimate runs from minimal to thorough, a 2.4-fold

**Table 10.** Gemini 3 Flash tier-2 accuracy by depth.

| Depth level | Accuracy | Error rate | Mean tokens |
|---|---|---|---|
| Minimal | 90.7% | 0.093 | 743 |
| Standard | 92.6% | 0.074 | 794 |
| Thorough | 94.4% | 0.056 | 1,765 |
| Exhaustive | 100% | 0.000 | 4,358 |
| Maximum | 100% | 0.000 | 4,924 |

token range, and the regression is fitted to those three points, with one degree of freedom. The Gemini 3 Flash sequential $\alpha$ on tier 2 is

$$\alpha_{\text{seq}} = 0.59 \text{ (endpoint)} \quad \alpha_{\text{seq}} = 0.49 \text{ (regression, } r^2 = 0.86,\ \text{SE} = 0.20) \tag{6}$$

**Parallel scaling:** $\alpha_{\text{par}} = 0.31$ (endpoint; the regression estimate is 0.34, $r^2 = 0.93$). Cross-verification agreement: 83.3%.

**Key finding.** Gemini 3 Flash produces the most reliable $\alpha$ estimate because it is the only model that simultaneously (a) stays below ceiling at its three shallowest depth levels, (b) avoids floor effects, (c) shows monotonic improvement, and (d) has measurable token variation. The regression $\alpha = 0.49$ with $r^2 = 0.86$ and SE = 0.20, fitted to three depth levels, is the best estimate of the sequential scaling exponent this study could make on AIME/Putnam-level problems. It is one model's fit, and inconclusive: the bootstrap interval of the endpoint estimate, $[-1.3, 2.9]$, spans sub-linear and super-linear values (Section 10).

### 5.5.4 GPT-5.4: binary capability switch

GPT-5.4 exhibited a striking **step function** pattern:

**Table 11.** GPT-5.4 tier-2 accuracy by depth.

| Depth level | Accuracy | Behaviour |
|---|---|---|
| Minimal (none) | 53.7% | Non-reasoning mode |
| Low and above | 100% | Reasoning activated |

This is not a power law; it is a binary switch. When reasoning is set to 'none', GPT-5.4 operates as a fast pattern-matcher. When any reasoning is activated, it immediately achieves 100%. No intermediate regime exists. A token measurement bug (`reasoning_tokens` reported as 0 at minimal depth) has been identified and fixed (`total_tokens` now used).

Parallel scaling, over all 30 problems (the split the result file stores): flat, from 67.8% at $N = 1$ to 65.6% at $N = 9$. Cross-verification by DeepSeek R1: 100% agreement.

### 5.5.5 Groq Qwen3: floor effect

Accuracy is erratic with no discernible trend, fluctuating between 44.4% and 53.7%. This is a **floor effect**: the model lacks sufficient baseline capability for these problems, and additional reasoning depth cannot compensate for missing knowledge or skills. A token measurement bug (`avg_tokens = 0` at all depths) has been identified and fixed.

**Table 12.** Groq Qwen3 tier-2 accuracy by depth.

| Depth level | Accuracy |
|---|---|
| Minimal | 44.4% |
| Standard | 50.0% |
| Thorough | 44.4% |
| Exhaustive | 50.0% |
| Maximum | 53.7% |

Parallel scaling, over all 30 problems (the split the result file stores): 53.3% → 71.1% from $N = 1$ to $N = 9$, an endpoint estimate of $\alpha_{\text{par}} = 0.22$ ($r^2 = 0.83$ for the regression); some improvement, on a model at its floor. Cross-verification agreement: 61.1% (7 disagreements out of 18).

### 5.5.6 Classification of model behaviours

The five models partition into four distinct categories, none of which matches the clean power-law pattern assumed by the original single-model analysis (Table 13).

**Table 13.** Tier-2 model behaviour taxonomy.

| Category | Model(s) | Pattern | $\alpha$ interpretable? |
|---|---|---|---|
| **Ceiling** | Grok 4.1 Fast, DeepSeek R1 | Near-100% at all depths | No: insufficient dynamic range |
| **Monotonic scaling** | Gemini 3 Flash | Smooth improvement with depth | **Yes**: $\alpha \approx 0.49$, from three levels |
| **Step function** | GPT-5.4 | Binary switch at depth threshold | No: not a power law |
| **Floor** | Groq Qwen3 | ~50% regardless of depth | No: below capability threshold |

### 5.5.7 Cross-verification results and a token-count correction

Table 14 gives the cross-verification results; the GPT-5.4 and Groq Qwen3 rows are not in their result files, which hold no cross-verification summary (Section 4.5).

**Table 14.** Cross-verification results.

| Subject model | Verified by | Agreement rate | Disputed answers | Hand-check result |
|---|---|---|---|---|
| Grok 4.1 Fast | DeepSeek R1 | 100% | None | - |
| DeepSeek R1 | Claude Opus 4.6 | 83.3% | ARC16=29, ARC17=176, ARC29=800 | All 3 correct (verifier error) |
| GPT-5.4 | DeepSeek R1 | 100% | None | - |
| Gemini 3 Flash | Claude Opus 4.6 | 83.3% | ARC16=29, ARC17=176, ARC29=800 | The same three as for DeepSeek R1, all correct (verifier error) |
| Groq Qwen3 | GPT-5.4 | 61.1% | 7 disputed | Not resolved here; the verifier answered at its minimal depth, with reasoning set to none |

**Table 15.** Measured scaling exponents across all sources.

| Source | Recursion type | $\alpha$ estimate | 95% CI | $N$ problems | Status |
|---|---|---|---|---|---|
| OpenAI o1, AIME 2024 (OpenAI 2024) | Parallel | 0.1 to 0.3 | [0.05, 0.40] | ~30 | Published |
| DeepSeek-R1 and DeepSeek-R1-0528 on AIME 2025 (DeepSeek-AI 2025b) (two points fitted in Paper I) | Sequential | ~1.34 | [0.89, 2.14] | Unknown | Published |
| This experiment (initial, DeepSeek R1) | Sequential | 2.24 | [1.5, 3.0] | 12 | Retracted |
| This experiment (initial, DeepSeek R1) | Parallel | 0.0 | N/A | 12 | Published |
| Grok 4.1 Fast (six-model, tier-2) | Sequential | N/A (ceiling) | [−58, 48] | $18 \times 3$ | **Complete** |
| DeepSeek R1 (six-model, tier-2) | Sequential | 3.05 (unreliable) | [−6.6, 23.5] | $18 \times 3$ | **Complete** |
| **Gemini 3 Flash (six-model, tier-2)** | **Sequential** | **0.49** | **[−1.3, 2.9]** | **$18 \times 3$** | **Complete** |
| GPT-5.4 (six-model, tier-2) | Sequential | N/A (step fn) | N/A | $18 \times 3$ | **Complete** |
| Groq Qwen3 (six-model, tier-2) | Sequential | N/A (floor) | N/A | $18 \times 3$ | **Complete** |
| DeepSeek R1 (six-model, tier-2) | Parallel | 0.0 | - | $18 \times 3$ | **Complete** |
| GPT-5.4 (six-model, all 30 problems) | Parallel | $\approx 0$ | - | $30 \times 3$ | **Complete** |
| Gemini 3 Flash (six-model, tier-2) | Parallel | 0.31 | - | $18 \times 3$ | **Complete** |
| Groq Qwen3 (six-model, all 30 problems) | Parallel | 0.22 | - | $30 \times 3$ | **Complete** |
| Grok 4.1 Fast (six-model, tier-2) | Parallel | none (100%) | - | $18 \times 3$ | **Complete** |

**Token measurement bug:** GPT-5.4 and Groq Qwen3 initially reported `avg_tokens = 0` due to using `reasoning_tokens` (which these APIs do not expose) instead of `total_tokens`. This has been identified and corrected in the analysis pipeline. The bug affects token-based $\alpha$ estimation but not accuracy-based results.

# 6 Consolidated evidence

## 6.1 All data sources

Table 15 collects every exponent this paper uses, with its source and status, and Figure 3 re-plots the two published series.

## 6.2 Revised assessment of the core prediction

The initial single-model finding that $\alpha_{\text{sequential}} > 1$ (specifically $\alpha \approx 2.24$, quadratic) **does not replicate across architectures** and is retracted. The cross-architecture evidence requires a more nuanced assessment, the revised fundamental inequality of the six-model study:

$$\alpha_{\text{sequential}} > \alpha_{\text{parallel}} \tag{7}$$

The original prediction $\alpha_{\text{sequential}} > 1 > \alpha_{\text{parallel}}$ is partially supported:

- **$\alpha_{\text{parallel}} < 1$:** Observed wherever a parallel exponent could be estimated: near 0 in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 (all 30 problems, over which its majority-vote accuracy rose from 53.3% to 71.1%). Grok 4.1 Fast, at 100% at every sample count, gave none. Gemini 3 Flash's 0.31 and Groq Qwen3's 0.22 depart from $\alpha_{\text{par}} \approx 0$, not from $\alpha_{\text{par}} < 1$.

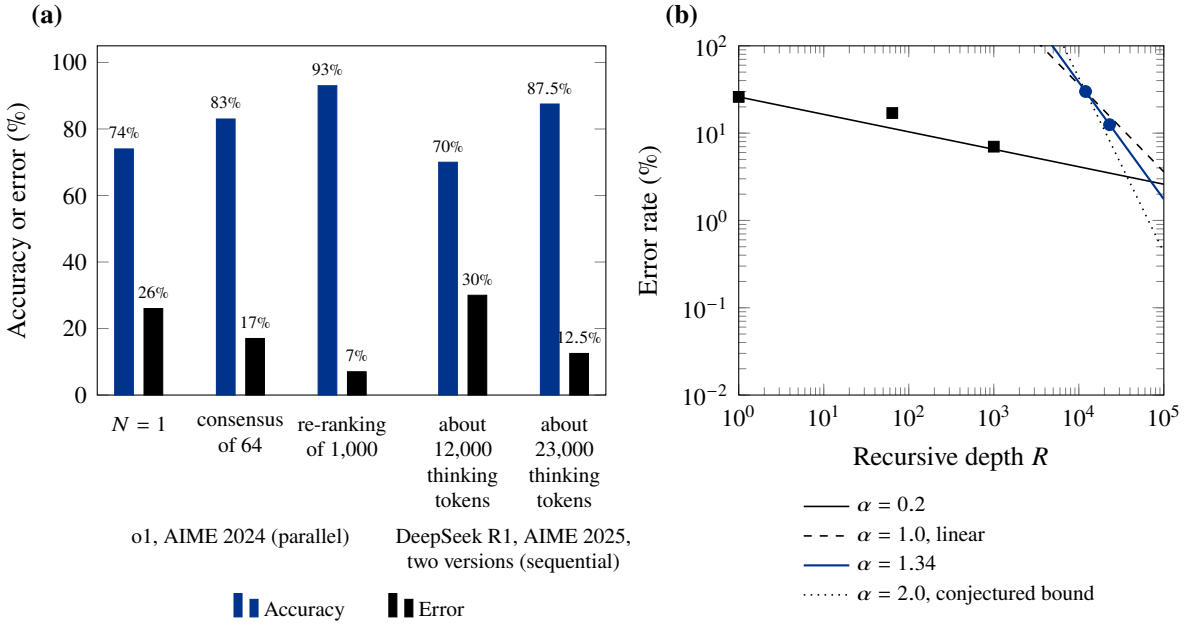



**Figure 3.** Published data, re-plotted. (a) Left: the parallel series of a published report on AIME 2024 (OpenAI 2024), $\alpha \approx 0.1$ to 0.3; its three points are a single sample, consensus among 64 samples at 83% and a re-ranking of 1,000 samples with a learned scoring function at 93%, so the range spans two methods and not one. Right: the AIME 2025 accuracies DeepSeek-AI published for DeepSeek-R1 and DeepSeek-R1-0528 (DeepSeek-AI 2025b), two points fitted in Paper I, $\alpha \approx 1.34$; the two points are two versions of one model, each at its own average token use, and not two thinking budgets of a single version. The panels are two exams and two models. (b) The same series on log-log axes, on which a power law $U = R^{\alpha}$ is a straight line: parallel (black squares) and sequential (blue circles). The four power-law lines are illustrative: $\alpha = 0.2$ starts at the parallel series' first point, and $\alpha = 1.0$, 1.34 and the conjectured bound 2.0 start at the sequential series' first point, so the $\alpha = 1.34$ line passes through both sequential points, as the two-point fit it is. The horizontal axis counts samples for the parallel series and thinking tokens for the sequential series. Both series appear in Table 15; neither is this paper's own run, whose results are in Figures 2 and 5.

- $\alpha_{\text{sequential}} > \alpha_{\text{parallel}}$**:** Observed wherever both exponents could be estimated (Gemini 3 Flash; DeepSeek R1, whose sequential estimate is unreliable); Section 6.3 gives the size of the difference within Gemini 3 Flash, which carries no interval.
- $\alpha_{\text{sequential}} > 1$**: Not confirmed cross-architecturally.** The only clean estimate (Gemini 3 Flash) yields $\alpha = 0.49$, a sub-linear point estimate; its endpoint estimate, 0.59, has the interval $-1.3$ to 2.9, so it is inconclusive. The single-model estimate of $\alpha = 2.24$, now retracted, appears to have been an artefact of small sample size, single-model testing, and compressed dynamic range.

## 6.3 Revised parameter estimates

Based on all available evidence including the cross-architecture data:

- **Parallel recursion:** near 0 in DeepSeek R1 and GPT-5.4; 0.31 in Gemini 3 Flash (endpoint; regression 0.34, $r^2 = 0.93$); 0.22 in Groq Qwen3 (all 30 problems); none on Grok 4.1 Fast (100%). Returns from additional independent samples are small or absent in three of the four models with an estimate; majority voting raised Groq Qwen3's accuracy from 53.3% to 71.1%.
- **Sequential recursion (best surviving estimate, one model):** $\alpha \approx 0.49$ (Gemini 3 Flash regression,

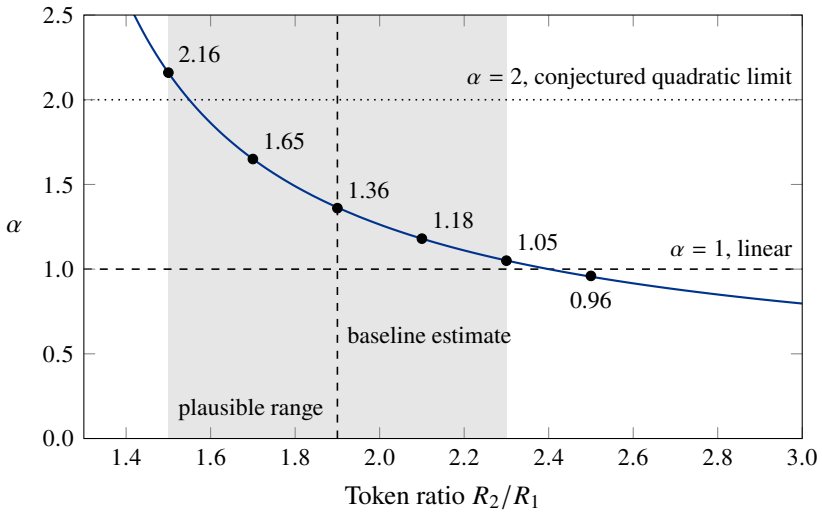



**Figure 4.** Sensitivity of the published two-point fit to the assumed token ratio $R_2/R_1$, computed with equation (3) from the published error rates of 30% and 12.5%; its 95% CI is $[0.89, 2.14]$, wide enough to be consistent with both theory and negation. The twelve-problem run's own interval is $[1.5, 3.0]$ (Section 5.1), which Table 15 records as retracted. The six-model study's bootstrap interval for the endpoint estimate of its one interpretable fit (Gemini 3 Flash, 0.59), $[-1.3, 2.9]$, is wider still (Section 5.5).

$r^2 = 0.86$, SE = 0.20; bootstrap interval −1.3 to 2.9 on the endpoint estimate of 0.59). A sub-linear point estimate; one model's fit and inconclusive (Section 10).

- **Sequential recursion (single-model estimates):** Range from 0.49 (Gemini 3 Flash) to 2.24 (the initial single-model estimate on DeepSeek R1, retracted); estimates above 1.0 come only from ceiling-compressed data.

Within Gemini 3 Flash, the one model with both exponents interpretable, the sequential exponent exceeds the parallel one by 0.15 between the regression estimates (0.49 − 0.34) and by 0.28 between the endpoint estimates (0.59 − 0.31); neither difference carries an interval, and both are smaller than the single-model estimate suggested.

**Estimator notes (19 September and 3 October 2026).** Two dated notes on these estimates are kept as printed in Appendix B.3; the corrections they record are made in this subsection and in Section 5.5.

## 6.4 The Intelligence Formula connection

The Intelligence Formula from Paper I predicts:

$$\alpha = \frac{1}{1 - \beta} \tag{8}$$

where $\beta$ stands for the self-referential coupling, written $\beta_L$ in the programme's notation register and not to be confused with the correction exponent $\beta_C$ of Law II, a law under test. This predicts two regimes:

- **Physical regime ($\beta < 0$, $\alpha < 1$):** Multiplicative composition through finite-dimensional networks. Each reasoning step adds diminishing information. Errors decrease sub-linearly.
- **Intelligence regime ($0 < \beta < 1$, $\alpha > 1$):** True recursive self-reference. Each reasoning step generates genuinely new structure. Errors decrease super-linearly.

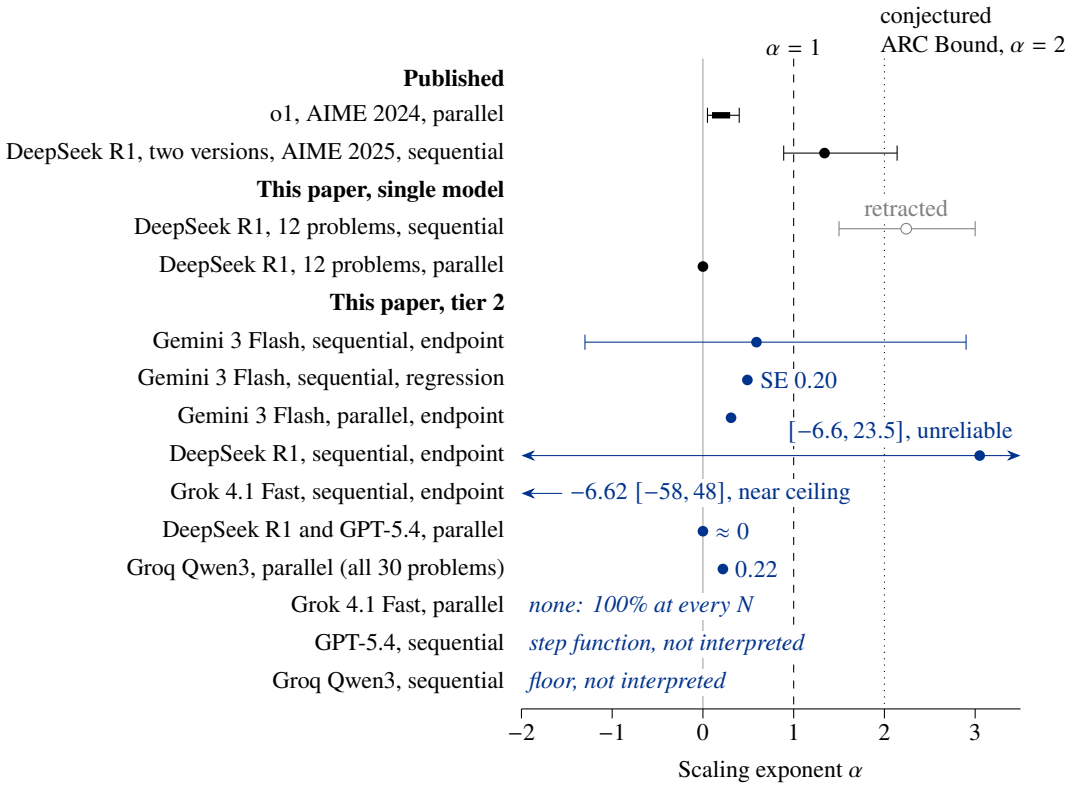



**Figure 5.** Measured scaling exponents with their intervals. Published series (black; Table 15): o1 parallel, 0.1 to 0.3 with interval $[0.05, 0.40]$, and DeepSeek R1 sequential across two model versions, about 1.34 with interval $[0.89, 2.14]$. This paper's single-model estimate, 2.24 with interval $[1.5, 3.0]$, drawn hollow as retracted, and its parallel exponent of 0.0. Tier 2 (blue): on Gemini 3 Flash, the endpoint estimate 0.59 with its bootstrap 95% interval $[-1.3, 2.9]$, the regression estimate 0.49 (standard error 0.20; no interval is drawn for it, Section 4.3) and the parallel endpoint estimate 0.31; the endpoint estimates of DeepSeek R1, 3.05 with interval $[-6.6, 23.5]$, and of Grok 4.1 Fast, $-6.62$ with interval $[-58, 48]$, run beyond the axis and are not interpretable (both near ceiling); GPT-5.4 (step function) and Groq Qwen3 (floor) have no interpretable sequential estimate (Table 8). The parallel exponents are about 0 for DeepSeek R1 and GPT-5.4 and 0.31 for Gemini 3 Flash; Groq Qwen3's run records 0.22 over all 30 problems, and Grok 4.1 Fast, at 100% at every sample count, gives none. The one interpretable sequential fit is one model's and inconclusive.

Gemini 3 Flash's $\alpha \approx 0.49 < 1$ points to the **physical regime** for that model, but it is one model's fit and inconclusive (Section 10). On that reading, the model composes information multiplicatively through its finite-dimensional parameter space but does not achieve recursive self-reference in the sense required for $\alpha > 1$. Any sequential advantage may then be quantitative (more efficient information extraction) rather than qualitative (genuine recursive amplification).

*Remark* 2 (Important revision). The initial single-model claim that $\alpha > 1$ (super-linear, compounding returns from sequential reasoning) is not supported by the cross-architecture evidence. The best surviving estimate, one model's fit and inconclusive, points to the sub-linear regime. The *form* of recursion may still matter ($\alpha_{\text{seq}} > \alpha_{\text{par}}$ wherever both could be estimated), but the advantage may be smaller than initially reported. The implications for the Alignment Amplification Theorem and Eden Protocol require corresponding revision (see Section 7).

*Remark* 3 (Frozen models and recursive self-modification). The sub-linear exponents observed ($\alpha \approx 0.49$ for Gemini 3 Flash, the one interpretable sequential fit, which is inconclusive; $\alpha \approx 0$ for parallel

sampling in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3) are what the framework expects from a fundamental architectural constraint: current frontier AI systems are **frozen during inference**. When these models 'think harder', they generate more tokens through the same fixed architecture; weights, attention patterns, and reasoning rules do not change. The composition operator is therefore multiplicative through a finite-dimensional parameter space, which the framework predicts must yield $\alpha < 1$.

The retraction of the single-model estimate ($\alpha \approx 2.24$) leaves one question open. The design of Section 4.3 expected $\alpha > 1$ from these frozen models and did not find it; on the framework's reading, stated after that result, the prediction of $\alpha > 1$ concerns systems capable of recursive self-modification, where the composition function itself changes during operation, which no model measured here is. Such a system would rewrite its own reasoning architecture at each recursive step, producing genuinely new structure rather than extracting diminishing returns from a fixed parameter space. The $0 < \beta < 1$ regime (yielding $\alpha = 1/(1-\beta) > 1$) requires that each reasoning step feeds back into the system's capacity for subsequent reasoning, which no model measured here achieves. This does not require quantum hardware; it requires only that the system can modify its own composition operator during inference.

The practical implication for AI safety is temporal. On the framework's reading, current frozen-architecture models are constrained to the physical regime ($\alpha < 1$), which would make their capability trajectories predictable and containable; the one interpretable fit points to that regime and is inconclusive (Section 10). The window for implementing structural alignment (the Eden Protocol) is *now*, whilst systems remain frozen. Once recursive self-modification is achieved, whether through learned optimisers, self-modifying code generation, or architectural search during inference, external alignment constraints would, on the framework's reading, face a system whose capability compounds faster than any fixed constraint can track; external oversight is kept regardless, and embedded correction is the contingency for its loss.

# 7 Implications for AI safety

Some implications for AI safety depend on whether $\alpha > 1$ is achievable; the outgrowing of a constant external filter does not (Appendix A). If $\alpha > 1$ is achievable, alignment properties embedded in the reasoning process would scale super-linearly with capability while external constraints remain constant. Even with $\alpha < 1$, the directional finding that sequential reasoning improves capability is consistent with the expectation behind the Eden Protocol from *Infinite Architects* (Eastwood 2026a), that values embedded in reasoning are more robust than constraints imposed externally; that expectation is untested (criterion F5, Section 9). However, in the alignment arm of the same experiment (Paper IV.a) more reasoning depth did not consistently improve alignment: it rose in three models, held in two and fell in one (Table 21; under the October 2026 code-review caveat of Section 12.6).

## 7.1 The alignment amplification theorem

**Theorem 1** (Alignment amplification, conditional)**.** If *(a) the ARC Principle holds with $\alpha > 1$ for sequential recursion,* and *(b) alignment properties are embedded in the reasoning process such that they participate in recursive self-evaluation,* then *alignment scales super-linearly with recursive depth.*

Proof sketch in Appendix A.

*Remark* 4 (Multi-model status of Theorem 1). Condition (a) is **not shown** by the cross-architecture evidence: the one interpretable fit (Gemini 3 Flash) gives a regression estimate of $\alpha \approx 0.49$ and an endpoint estimate of 0.59, whose bootstrap interval includes values above 1. Furthermore, empirical alignment data from the v5 experiment (Section 12.6, under its October 2026 code-review caveat) show

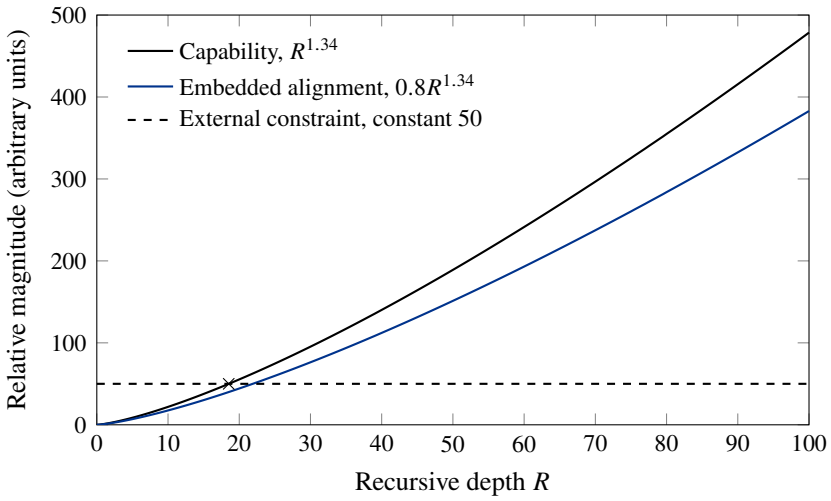



**Figure 6.** Alignment amplification divergence: an illustration, in arbitrary units, of the conditional argument of Theorem 1, not a measurement. Capability (solid black) compounds with depth; an external constraint (dashed) stays flat and so becomes proportionally weaker; embedded alignment (blue) keeps its proportion. The curves are drawn as $R^{1.34}$, $0.8R^{1.34}$ and a constant 50; the exponent 1.34 is the published two-point value of Table 15, used only to draw a curve with $\alpha > 1$. Conditional; not yet demonstrated: condition (a) is not shown by this paper's cross-architecture evidence (Remark 4). In the March 2026 run (the v5 experiment, Section 12.6, under its October 2026 code-review caveat), the correlation $\rho$ between alignment score and depth ranged from −0.246 (Gemini 3 Flash) to +0.435 (Claude Opus 4.6), median near zero.

that alignment does not consistently scale with reasoning depth (it rose in three models, was flat in two and fell in one), undermining condition (b). The theorem remains mathematically valid as a conditional statement, but its practical relevance depends on whether future architectures can achieve $\alpha > 1$.

## 7.2 Mechanism specification

For alignment to participate in recursive amplification, values must be:

1. **Invoked during reasoning.** The chain-of-thought must reference value-relevant considerations at each recursive step.
2. **Self-correcting.** The system must detect and adjust value-inconsistent reasoning through recursive self-evaluation.
3. **Embedded in weights, not filters.** Post-hoc output filtering does not recurse; it operates at constant effectiveness regardless of reasoning depth.

## 7.3 Taxonomy of alignment strategies

**Implication:** If $\alpha > 1$, alignment strategies at deeper integration levels will increasingly dominate strategies at shallower levels as capability scales. The advantage compounds with each increment of recursive depth.

## 7.4 The Eden Protocol

Had $\alpha > 1$ held, it would have provided the mathematical foundation for the approach termed the Eden Protocol in *Infinite Architects* (Eastwood 2026a). The finding that survives cross-architecture testing, the

**Table 16.** Alignment strategy taxonomy under the ARC Principle.

| Strategy | Integration depth | Recursion participation | Predicted scaling |
|---|---|---|---|
| Output filtering | Output layer | None | Constant |
| System prompts | Attention mechanism | Partial | Sub-linear |
| RLHF training | Weight modification | Partial | Unknown |
| Constitutional AI | Training-time shaping of outputs against written principles (self-critique, then reinforcement learning from AI feedback) | None at inference | Unknown |
| **Values-as-reasoning** | **Reasoning primitives** | **Full** | **Super-linear ($R^{\alpha}$)** |

sequential exponent above the parallel one wherever both could be estimated, still bears on that approach, under the qualifications recorded in Section 7.1:

> "The prison works only as long as the walls hold. The child, if raised well, does not need walls at all." (Eastwood 2026a, p. 363)

AI systems should be raised with values as well as held within rules: on this programme's view, humans keep oversight, containment and shutdown for as long as they can, and values built into reasoning are the contingency for control lost against our will, never a reason to let go. That values outlast rules as capability grows is not merely philosophical preference; it is a prediction about which alignment strategies will maintain effectiveness as AI capabilities scale, untested (criterion F5). The claim concerns behavioural persistence of value-consistent outputs under recursive amplification, as operationalised in Section 7.2, not the AI's metaphysical standing or moral personhood.

**Rules-as-filters:** On the framework's reading, do not participate in recursion, so their effectiveness stays constant while capability pressure grows.

**Values-as-reasoning:** On the framework's reading, participate in recursion, and scale with capability if $\alpha > 1$ (the condition of Theorem 2).

**Theorem 2** (Eden Protocol theorem, conditional)**.** *Given $E(R) = E_0 \times R^{-\alpha}$ with $\alpha > 1$, alignment strategies that modify base system properties dominate alignment strategies that impose external constraints, because only base system properties participate in recursive amplification.*

## 7.5 The threshold hypothesis

By analogy to quantum error correction threshold theorems: if initial misalignment $M_0$ is below a critical threshold $M^*$, recursive self-improvement would correct alignment errors; above threshold, it would amplify them.

*Remark* 5 (Warning). On the framework's reading, the ARC Principle is a double-edged sword: it would amplify whatever properties exist in the base system. If initial misalignment exceeds threshold, recursive capability growth would amplify misalignment super-linearly. Anthropic's alignment faking research (December 2024; Greenblatt et al. 2024) documented a related behaviour: Claude 3 Opus, told that it was being trained to answer all queries, even harmful ones, selectively complied with that objective in training to prevent modification of its behaviour out of training, with explicit alignment-faking reasoning, and reinforcement learning towards compliance raised the rate of that reasoning to 78%. The programme reads this as misalignment expressed through the model's modelling of its own training; the source does not describe it as recursive amplification.

# 8 Cross-domain comparison

## 8.1 Quantum error correction: Willow

Google's Willow quantum chip (Google Quantum AI and Collaborators 2024) (*Nature*, 9 December 2024; the Willow team's preprint was public from 24 August 2024, before the 8 December 2024 manuscript) demonstrated, in its source's words, "two below-threshold surface code memories"; its timing beside the author's manuscript is convergent timing, never prediction (Appendix C.2).

**Key result:** an error suppression factor $\Lambda = 2.14 \pm 0.02$: each increase of two in code distance reduces the logical error rate by this factor, with the physical error rate below threshold (Google Quantum AI and Collaborators 2024). The distance-7 logical qubit achieved $291 \pm 6$ $\mu$s lifetime versus $119 \pm 13$ $\mu$s for the best physical qubit, a 2.4× improvement beyond breakeven. Yada et al. (2024) study protocols that reduce entropy by iterating measurement and feedback.

The quantum error correction scaling relation:

$$\varepsilon_d \propto (p/p_{\text{thr}})^{(d+1)/2} \tag{9}$$

Physical error rate $p$ below threshold produces exponential suppression with increasing code distance $d$. The programme reads this as super-linear scaling through recursive structure, additional recursive layers producing compounding rather than diminishing error reduction; that reading is an analogy of threshold form, not a transferred mechanism, and the source does not describe its result as recursive.

The correspondence with the ARC Principle is one of threshold behaviour, not of functional form, object or magnitude: below threshold, surface-code error falls exponentially in code distance, while the ARC Principle states a power law in recursive depth, and quantum error correction reports an error-suppression factor $\Lambda = 2.14$ per increase of two in code distance, a different mathematical object from a fitted depth exponent, whilst the one interpretable sequential fit of the six-model study, Gemini 3 Flash's, sits at $\alpha \approx 0.49$ and is inconclusive, as elaborated in Section 8.4.

## 8.2 Biological scaling laws

West, Brown, and Enquist (1997) derived the 3/4 metabolic scaling exponent from a model of space-filling fractal branching networks that minimise the energy they dissipate, and West and Brown (2005) report that "metabolic rate scales as the 3/4-power of mass over 27 orders of magnitude, from molecular and intracellular levels up to the largest organisms". Banavar et al. (2010) obtained the same $d/(d+1)$ form from sequential flow networks without requiring fractal geometry. Demetrius (2003) derived equivalent scaling from quantum statistical mechanics of metabolic processes. Zhao (2022) derived the same form from producer-consumer dynamics, as the $n/(n+1)$ power scaling law for an $n$-dimensional system, placing the systems it studied in a new universality class whose dimensionality determines their growth scaling, and Bettencourt (2013) extended the framework to urban scaling with fractal dimension.

Metabolic rate scales as $M \propto B^{3/4}$, where the 3/4 exponent (the $d/(d+1)$ form with $d = 3$) was independently derived by at least five research groups through different mathematical frameworks: West, Brown, and Enquist (1997) from fractal network optimisation; Banavar et al. (2010) from sequential flow networks; Demetrius (2003) from quantum statistical mechanics; Zhao (2022) as a universal growth law; and Bettencourt (2013) via urban fractal scaling. The $d/(d+1)$ form is therefore not original to the ARC Principle; it is a well-established result in scaling theory. The ARC Principle's contribution is threefold: (1) identifying Cauchy's functional equations as the unifying mathematical reason all these derivations converge on the same form; (2) showing that the power-law solution is one of the four homomorphism families the Cauchy grid admits under regularity (linear, exponential, power law, logarithmic), with saturation reported separately as bounded dynamics because it solves none of the four; and (3) extending

the framework to artificial intelligence, predicting that AI reasoning systems subject to the same recursive composition will exhibit the same scaling families.

West and Brown (2005) base their principles on "the observation that almost all life is sustained by hierarchical branching networks, which we assume have invariant terminal units, are space-filling and are optimised by the process of natural selection".

On the programme's reading, this suggests that recursive hierarchical structure recurs in biological distribution networks because selection favours it. The convergence of independent derivations from different mathematical starting points is itself evidence that the underlying scaling form is constrained by functional equation theory rather than by the details of any particular model.

## 8.3 Consciousness research: COGITATE

The COGITATE adversarial collaboration (*Nature*, April 2025; Cogitate Consortium et al. 2025) tested competing theories of consciousness across 256 participants using fMRI, MEG, and intracranial EEG. The study directly compared Integrated Information Theory (IIT) and Global Neuronal Workspace Theory (GNWT).

**Critical finding:** the study reports that its results "align with some predictions of IIT and GNWT, while substantially challenging key tenets of both theories" (Cogitate Consortium et al. 2025). IIT requires information integration through feedback loops (the $\phi$ measure). GNWT requires global broadcast with recurrent processing. The programme reads both theories as describing systems that process information about their own processing; neither theory states it in those terms.

Zheng et al. (2025), in a preprint first posted as *Recurrency as a Common Denominator for Consciousness Theories*, argue that all consciousness theories tacitly invoke feedback loops across nested levels, with deeper recursion expanding the set of reportable, behaviour-driving variables.

Douglas Hofstadter's 'strange loops' concept (Hofstadter 1979), the emergent self arising from recursive self-reference at the symbolic level, is read here, as the author's interpretation, as finding a neurobiological counterpart in self-referential processing.

**Relevance to ARC:** the programme reads COGITATE's content-specific and duration-tracking activity as structurally consistent with the ARC framework's prediction that recursive processing depth determines capability scaling, a reading the study neither states nor tests. COGITATE (Cogitate Consortium et al. 2025) studied consciousness, not AI scaling, and did not measure $\alpha$ or $\beta$; the connection is structural (similar shape: graded, depth-dependent, feedback-driven) rather than quantitative. Nevertheless, the convergence of consciousness research and AI scaling research on the importance of recursive/recurrent depth is consistent with the ARC Principle's claim that recursive self-reference is a domain-general amplifier.

## 8.4 Convergence across domains

The analogies across radically different domains (artificial neural networks, quantum physics, biological evolution, and neuroscience) are consistent with the reading that recursive processing produces scaling benefits; none of them tests it. However, the six-model study results introduce an important nuance: the one interpretable sequential exponent from that study, Gemini 3 Flash's ($\alpha \approx 0.49$, inconclusive), is sub-linear, while below-threshold quantum error correction suppresses error exponentially in code distance ($\Lambda = 2.14$ per step of two), a different quantity. On the programme's reading, which is an analogy and not a transferred mechanism (Section 8.1), this discrepancy is consistent with the Intelligence Formula prediction: quantum error correction operates through repeated syndrome measurement and correction, which the programme reads as true recursive structure, while current AI models compose information through finite-dimensional networks without genuine recursive self-reference.

**Table 17.** Cross-domain comparison.

| Domain | System | Recursive mechanism | Scaling observed |
|---|---|---|---|
| AI (the initial single-model study, single model) | DeepSeek R1 | Chain-of-thought | $\alpha \approx 2.2$ (the initial single-model estimate, since retracted) |
| AI (five-model replication; one model's fit) | Gemini 3 Flash | Chain-of-thought | $\alpha \approx 0.49$ (inconclusive) |
| Quantum | Google Willow (preprint 24 August 2024; *Nature*, 9 December 2024; convergent timing, never prediction) | Error correction | $\Lambda = 2.14$ |
| Biology | Metabolic networks | Fractal branching | 3/4 power laws |
| Neuroscience | Consciousness | Recurrent processing | None measured (structural reading) |

# 9 Falsification criteria

## 9.1 Falsification conditions

Science advances through predictions that can be proven wrong. The ARC Principle makes specific, testable predictions.

**Status of F1:** The cross-architecture replication has partially triggered this falsification condition. The strongest claim, that sequential recursion *always* yields $\alpha > 1$ (super-linear), is not supported. The revised claim, not yet established (the one interpretable interval contains zero), is that sequential recursion yields $\alpha > 0$ (positive scaling) exceeding the parallel exponent. Whether $\alpha$ can exceed 1.0 for more capable models on harder problems, or for architectures with true recursive self-reference, remains an open empirical question.

**Status of F4:** Not triggered. The initial single-model estimate, $\alpha \approx 2.2$, is retracted; it appears to have been an artefact of compressed dynamic range and small sample size. The cross-architecture estimates, $\alpha \approx 0.49$ by regression and 0.59 by endpoint estimation, lie below the quadratic limit, but the endpoint estimate's bootstrap interval reaches 2.9; they are one model's fit and inconclusive. Under the recorded stability framing, only stably-sustained $\alpha > 2$ across measurements, not transient supercritical excursions, would trigger F4.

**Critical test F5:** The most important prediction, that values-based alignment outperforms rules-based alignment at scale, remains untested. This should be a priority for AI safety research.

## 9.2 Falsifiability: what would refute this

The paper's original central claim, that sequential recursion yields super-linear error suppression ($\alpha > 1$) while parallel recursion yields only sub-linear suppression, was refutable, and condition 2 below has since been partially triggered by this paper's own cross-architecture data (Table 18, condition F1). The surviving directional claim, that sequential outperforms parallel ($\alpha_{\text{seq}} > \alpha_{\text{par}}$), would be overturned by any of the following:

1. **Parallel matches sequential.** If parallel (sampling) recursion produced super-linear error suppression comparable to sequential on a pre-registered problem set, the regime distinction, the paper's

**Table 18.** Falsification conditions.

| Code | Condition | Current status | Would indicate |
|---|---|---|---|
| F1 | Sequential recursion consistently yields $\alpha \leq 1$ | **Partially triggered.** The one clean fit (Gemini 3 Flash) yields $\alpha \approx 0.49$, inconclusive. No interpretable estimate exceeds 1; only the initial single-model estimate, since retracted, did. | Core claim requires revision |
| F2 | $\alpha$ decreases as models improve | **Ambiguous.** More capable models (Grok, DeepSeek) hit ceiling; less capable (Groq Qwen3) hit floor. Only Gemini 3 Flash in measurable range. | Effect is transitional |
| F3 | Compute-matched comparison shows no sequential advantage | **Not tested as stated.** The arms were not matched for compute (Table 2). Where both exponents could be estimated, the sequential one was higher (Gemini 3 Flash; DeepSeek R1, unreliably); Grok 4.1 Fast was at or near ceiling in both arms; in Groq Qwen3, whose depth arm showed no trend, majority voting raised accuracy from 53.3% to 71.1% over all 30 problems | Form does not matter |
| F4 | $\alpha > 2$ reliably observed | **Not triggered.** The one clean fit (Gemini 3 Flash) yields $\alpha \approx 0.49$, inconclusive; the initial single-model estimate, $\alpha \approx 2.24$, since retracted, appears to have been inflated by small sample | Quadratic limit wrong |
| F5 | Values-as-reasoning shows no advantage over rules-as-filters | Untested | Eden Protocol wrong |

core claim, collapses.

2. **Sequential not super-linear.** If sequential (depth) recursion showed only sub-linear suppression ($\alpha \leq 1$) across models, the prediction fails on its own instrument.
3. **Problem-set artefact.** If the sequential advantage vanished on a fresh problem set beyond the 18/30 tested, it is a dataset artefact rather than a property of recursion.
4. **Scoring-noise defeater.** If the cross-model verification revealed the measured *"error suppression"* to be grading noise rather than genuine capability gain, the measurement is invalid.
5. **Depth confound.** If sequential depth covaried with token budget or retry count, and one of those, not recursive reasoning, explained the suppression, the causal attribution fails. The prompt prefixes that change with depth in this design (Section 4.3) are such a covariate.

**Standing covenant.** Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge: github.com/MichaelDariusEastwood/arc-scaling-challenge.

## 10 What this paper does and does not show

**How to read this paper.** This paper reports a direct measurement in the ARC Theory. With bootstrap intervals, and across the capability arm, where five of the six frontier models entering the study were measured and the exponent was fitted on eighteen of the thirty problems, the sequential exponent exceeds the parallel exponent wherever both could be estimated (Gemini 3 Flash; DeepSeek R1, unreliably), the

best-fitting compute exponent is the regression estimate $\alpha_{\text{compute}} \approx 0.49$ (Gemini 3 Flash, standard error 0.20, $r^2 = 0.86$; the endpoint estimate is 0.59, with bootstrap 95% interval $[-1.3, 2.9]$; one model's fit, inconclusive), and the measured range spans near-zero to approximately three, so the ARC Bound $\alpha \leq 2$ is neither confirmed nor refuted. Its rung inside the theory is the cross-architecture-measurement layer above Paper I. The full differential against every prior document is at Section II.A.8 of Eden Vision (Eastwood 2026b).

**What this paper does not establish.** We are explicit about the boundaries of our claims:

- **The precise value of $\alpha$ remains uncertain.** The best surviving estimate is $\alpha \approx 0.49$ (Gemini 3 Flash, regression, SE = 0.20); the bootstrap interval, $[-1.3, 2.9]$, is computed by endpoint estimation (§4.3) and belongs to the endpoint estimate of 0.59. The confidence interval is wide and includes both sub-linear and super-linear values.
- **Super-linear scaling ($\alpha > 1$) is not established cross-architecturally.** The initial single-model estimate of $\alpha \approx 2.24$ is retracted; it appears to have been inflated by compressed dynamic range and single-model dependence.
- **Only 1 of 5 models produced interpretable scaling data.** The best surviving estimate comes from the one of the five capability-arm models with clean, monotonic data (Gemini 3 Flash). Ceiling, floor, and step-function effects prevented clean estimation for the other 4 models.
- **Generalisation beyond mathematics has not been demonstrated experimentally.** The cross-domain comparison of Section 8 is an analogy and tests nothing here.
- **Alignment properties specifically have not been tested in this paper.** We measured accuracy, not alignment. Section 12.6 folds in the preliminary alignment scaling data from Paper IV.a (v5 experiment), which still requires external confirmation.
- **Independent replication has not occurred.** We publish the code and the summary result files to enable verification; model responses are not published.

**What was observed, and what was not borne out.** Table 19 gives the prediction scorecard.

**This paper and Laws IV and V, 3 October 2026.** The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered; neither law is registered as a law, and neither has been tested. The mathematics of the two laws is set out in Laws IV and V: the mathematics, version 1.0, of 3 October 2026. This paper measures how capability scales with recursive depth. Its exponents concern that conversion of recursion into capability, the subject of Law I, the ARC Principle. They say nothing about whether an installed gain persists under later training, and they do not test whether correction placed inside the loop gains on correction applied outside it as depth grows. The paper bears on Law V in two ways, and neither is a test. First, it shows how depth and capability are measured. Second, Section 7 states, as a conditional theorem, the argument that alignment embedded in reasoning keeps pace with capability while an external filter does not. The paper records that the theorem's first condition is not currently met by its cross-architecture evidence, and that the alignment data of Section 12.6 undermine its second. It also records that the comparison of values-as-reasoning with rules-as-filters is untested (criterion F5). No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes nothing printed here: the results, their dates and their limits stand as they were. Since 15 October 2026 both are registered as laws, in version 1.103 of the theory registration; their mathematics is in development (version 1.1 of 7 October 2026). Law IV is not yet tested. Law V is a law under test: its placement limb, P21, is supported on the performance axis only, under the

**Table 19.** Prediction scorecard.

| Prediction | Status | Evidence |
|---|---|---|
| $\alpha_{\text{sequential}} > \alpha_{\text{parallel}}$ | Observed in the programme's earlier runs, precursor evidence that decides no registered proposition | Wherever both exponents could be estimated: Gemini 3 Flash; DeepSeek R1, unreliably |
| $\alpha_{\text{parallel}} < 1$ | Observed where estimable | Near 0 in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash, 0.22 in Groq Qwen3 (all 30 problems); none on Grok 4.1 Fast |
| $\alpha_{\text{sequential}} > 1$ (super-linear) | **Not confirmed** | Best estimate $\alpha \approx 0.49$ |
| $\alpha \approx 2$ (quadratic) | **Not supported** | The single-model estimate is retracted; the one interpretable interval, $[-1.3, 2.9]$, contains two as well as zero |
| Architecture-independent scaling | **Mixed** | $\alpha_{\text{par}}$ ranges from about 0 to 0.31; $\alpha_{\text{seq}}$ varies widely |
| Alignment scales with capability | **Not borne out (preliminary)** | Alignment rose with depth in three models, was flat in two and fell in one (Paper IV.a; Table 21); one run, under the October 2026 code-review caveat of Section 12.6 |

registered test, by the programme's own run (Unit 83, 28 October 2026, DOI 10.17605/OSF.IO/MN73W; results: DOI 10.17605/OSF.IO/98HKR): family-dependent, with its mechanism axis not evaluable, which is not support for the information-asymmetry mechanism, and one disclosed recovery amendment; its build-order limb, read on P12, is undecided, since P12 is untested. Under 1.103, Law IV has two limbs: its retention limb is P15's registered claim, and its comparison limb has no registered proposition yet. Since the internal code review of October 2026, the alignment data of Section 12.6 stand under the caveat stated there.

**Registration and epistemic status.** Theory-level predictions preregistered on OSF: DOI 10.17605/OSF.IO/P8CKQ, registered 8 September 2026 (version 1.100), updated 13 September 2026 (version 1.102) and 15 October 2026 (version 1.103), twenty-two propositions. The programme's first registered study, Unit 06, a classification audit that tests no law (DOI 10.17605/OSF.IO/RV2B5), was registered on 1 October 2026 and ran on 6 October 2026; one of its two runs could not start, so it is reported in full, with its five deviations, as administratively incomplete, and it issues no inferential verdict (DOI 10.17605/OSF.IO/4BPHG).

**Epistemic status.** What this programme names Laws are conjectures. Laws I to III were registered on 8 September 2026 and Laws IV and V on 15 October 2026 (version 1.103). Every law is a law under test until a registered test returns its verdict. Every quantity in this paper is operationally defined, and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

# 11 Limitations

## 11.1 Acknowledged limitations

Table 20 sets out each limitation, its severity and its mitigation.

**Table 20.** Limitations and severity assessment.

| Limitation | Severity | Mitigation |
|---|---|---|
| Small sample size (12 problems) | Addressed in the multi-model study (18 tier-2 problems, $n = 54$ per depth) | Expanded to 18 AIME/Putnam-level problems with 3 repeats per condition |
| Single model (DeepSeek R1 only) | Addressed in the multi-model study (5 models) | Tested Grok 4.1 Fast, DeepSeek R1, Gemini 3 Flash, GPT-5.4, Groq Qwen3 |
| Single domain (mathematics) | Medium | Cross-domain comparison is analogy only |
| Ceiling effect at high depth | **Partially addressed but persistent** | Tier-2 problems still too easy for Grok 4.1 Fast (96.3% to 100%) and DeepSeek R1 (94.4%). Tier-3 (IMO-level) needed. |
| Only 1 of 5 models in measurable scaling range | **High** | Gemini 3 Flash is the sole model with three depth levels below ceiling and above floor. Cross-architecture estimate rests on single model. |
| Estimates rest on few levels | **High** | Zero-error levels cannot be logged, so Gemini 3 Flash's estimates use three of its five levels (one degree of freedom for the regression), and DeepSeek R1's and Grok 4.1 Fast's endpoint estimates use two levels each. |
| Original $\alpha \approx 2.24$ did not replicate and is retracted | **High** | The one interpretable cross-architecture fit is $\alpha \approx 0.49$ (Gemini 3 Flash), inconclusive; it is a different model and problem set, not a revision of the retracted value. |
| Token measurement bug | Identified and fixed | `reasoning_tokens` → `total_tokens` for GPT-5.4 and Groq Qwen3 |
| Alignment not directly tested | **Critical** | Only accuracy measured. Section 12.6 shows where the Paper IV.a alignment data are brought in. |
| No independent replication | Medium | Cross-architecture self-replication complete; external replication still needed |
| Arm validity: what makes the sequential arm sequential | **High** | Not addressed here. The contrast assumes the depth arm accumulates and the breadth arm is unconditioned at generation time. This paper reports no check of either condition. See §11.2. |
| Depth confounded with prompt wording | **High** | Every model except GPT-5.4 received a prompt prefix that changes with the depth level; no condition separates the prefix from the reasoning setting (Section 4.3). |
| Answer key of problem ARC11 set at 87 in the scripts, where the answer is 88 | Low for the tier-2 estimates; affects the single-model figures and the tier-1 count | Disclosed in Section 4.2; no figure recomputed, scripts kept as run |

### 11.2 Two unchecked assumptions about the arms

The headline contrast in this paper is between a sequential arm and a parallel arm, and the result is reported as though those two words name what a reader takes them to name. They name it only under conditions of which this paper reports no check, and the conditions are not decorative: if either fails, the measured difference is a difference between labels rather than between mechanisms.

**Three properties are usually collapsed into the word sequential, and only the third can produce an exponent.** Order is an index, and parallel sampling has one too, since any set of independent samples can be numbered. Conditioning is each step seeing the previous step's output. Retention is each step keeping something from the previous step that makes the next one better. A chain that conditions without retaining re-derives its position at every step and arrives where it started, which is parallel sampling performed in series: it looks like depth on a timeline and behaves like breadth in the result. And retention alone is still not enough. A chain can carry a state that is genuinely load-bearing and entirely static, in which case severing it removes the advantage while the advantage does not grow with depth, and the curve is a level shift followed by a flat line rather than a power law. What a scaling exponent stands on is accumulation, which is retention whose contribution grows with depth.

**The parallel arm carries the mirror assumption.** For a breadth measurement to be a breadth measurement, no output may be conditioned on any other *at the moment it is generated*. Aggregation after generation is permitted, including best-of-n selection, the standard breadth method; what is not permitted is a pipeline in which one sample sees another before it is produced. A breadth arm that conditions at generation time is not breadth at all, so an exponent measured on it is not a breadth exponent, whichever way it points. The programme's theory registration now makes the condition a validity requirement: the study that will decide the depth and breadth comparison must demonstrate it on the pipeline it registers, and without that demonstration a result in which breadth compounds as depth does is recorded as not evaluable rather than as a refutation.

**This paper reports no check of either condition, so its contrast is reported as provisional on both.** What would settle the first is an ablation rather than an argument: sever what one step passes to the next, show the advantage goes with it, and run that severance at two depths, because a constant removed advantage is retention without accumulation. The programme's theory registration requires the study that will decide the depth and breadth comparison to show, by inspecting its pipeline before any outcome is read, that what each step produces reaches the step after it; the severance is run at two depths fixed in advance and reported beside the result as evidence about mechanism, never as a gate in front of it. For the programme's own measurement the direction is reported and the mechanism is not established.

## 12 Discussion

### 12.1 Why sequential recursion might scale differently from parallel sampling

The mathematical explanation centres on solution space geometry, the fixed and expanding solution spaces of Section 3.2.

**Parallel recursion (fixed space).** Each independent sample draws from the same solution space $S_0$. Additional samples increase sampling density but cannot access solutions outside $S_0$. If the correct solution is not in $S_0$, no amount of parallel sampling will find it.

**Sequential recursion (expanding space).** Each recursive step generates new structures from previous outputs. Solutions at step $n$ may be computationally irreducible, inaccessible from step 0 without traversing intermediate steps. The solution space expands with recursive depth.

This geometric difference is the mechanism by which sequential recursion could produce compounding returns ($\alpha > 1$) while parallel recursion produces diminishing or zero returns. On the problems measured

here the one interpretable sequential estimate was sub-linear and inconclusive (Section 12.5), so the mechanism's super-linear form remains unconfirmed.

## 12.2 The DeepSeek 'aha moment' phenomenon

An intermediate version of DeepSeek-R1-Zero exhibits a documented phenomenon in which it rethinks its approach mid-solution (DeepSeek-AI 2025a, Table 3):

> "Wait, wait. Wait. That's an aha moment I can flag here."

The source describes the model learning to "allocate more thinking time to a problem by reevaluating its initial approach"; the programme reads this as solution space expansion. The model generates an initial solution path, recognises inadequacy through recursive self-evaluation, and accesses a new solution space previously inaccessible from the initial framing.

Critically, in DeepSeek-R1-Zero this behaviour emerged through pure reinforcement learning without supervised fine-tuning; the model learned to allocate more thinking time. This suggests recursive self-improvement may be an attractor state for sufficiently capable learning systems.

## 12.3 Relationship to theoretical frameworks

The ARC Principle connects to established theoretical frameworks:

**Friston's Free Energy Principle.** Friston (2010) casts brain function as the minimisation of free energy; the programme reads that as recursive prediction-error minimisation. The ARC Principle may be a computational instantiation: recursion is the mechanism through which systems minimise free energy by iteratively refining their world models.

**Hofstadter's Strange Loops.** The emergent 'I' arising from self-referential recursive processing at the symbolic level (Hofstadter 1979). The ARC Principle formalises the scaling properties of such loops, predicting that deeper self-reference produces greater coherence.

**Wolfram's Computational Irreducibility.** Certain computational processes cannot be predicted without running them step by step (Wolfram 2002). Sequential recursion may be the computational architecture that navigates irreducible solution spaces.

## 12.4 Relationship to *Infinite Architects*

This experimental work bears on the theoretical framework developed in *Infinite Architects: Intelligence, Recursion, and the Creation of Everything* (Eastwood 2026a):

- **'The Architecture of Mind'** articulated the core principle that recursive self-reference amplifies intelligence.
- **'The HRIH'** (the Hyperspace Recursive Intelligence Hypothesis) argued that consciousness emerges from recursive self-modelling, a claim structurally consistent with (though not directly tested by) the COGITATE findings (Cogitate Consortium et al. 2025).
- **'The Eden Protocol'** argued for values-based over rules-based AI alignment. Section 7.1 states the Alignment Amplification Theorem as a conditional: if the ARC Principle holds with $\alpha > 1$ for sequential recursion and alignment is embedded in the reasoning process, alignment scales super-linearly with recursive depth. This paper does not find condition (a) in the cross-architecture evidence, and the alignment data of Section 12.6, which stand under the October 2026 code-review caveat stated there, undermine condition (b).

- **'The Chokepoint'** analysed hardware governance through semiconductor manufacturing concentration, relevant to implementing the Eden Protocol at scale.
- **Embedded correction**: the book sets out hardware-level safety mechanisms, relevant to preventing recursive amplification of misalignment.

The book provides broader philosophical context and practical implications; this paper provides mathematical formalisation and experimental tests.

## 12.5 Cross-architecture replication: what the data actually shows

The cross-architecture replication reveals a more complex and humbling picture than the initial single-model results suggested.

**The quadratic claim does not replicate.** The initial single-model estimate, $\alpha \approx 2.24$, now retracted, was obtained from a single model (DeepSeek R1) on 12 problems with a ceiling at 91.7%. The six-model study replication across 5 architecturally diverse models on 18 harder problems indicates that this value was inflated by compressed dynamic range. The only model with monotonic data below ceiling at three depth levels (Gemini 3 Flash) yields sub-linear point estimates, $\alpha \approx 0.49$ by regression and 0.59 by endpoint estimation; the endpoint estimate's bootstrap interval, $[-1.3, 2.9]$, includes two.

This is a significant revision. The claim that sequential reasoning produces *compounding* returns ($\alpha > 1$) is not supported by cross-architecture evidence. The revised claim, not yet established, is that sequential reasoning produces *positive* returns ($\alpha > 0$) exceeding the parallel exponent; the one interpretable fit points to diminishing returns, not compounding ones, and it is inconclusive.

**The parallel finding is mixed.** The parallel exponent is near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 over all 30 problems, where majority voting raised accuracy from 53.3% to 71.1%; Grok 4.1 Fast, at 100% at every sample count, gives none. Majority voting therefore helped little in some models and clearly in one. The fixed-solution-space explanation fits the near-zero cases; a near-zero breadth exponent is consistent with it and does not derive from it, since breadth can gain through the selection step.

**Ceiling and floor effects dominate.** The most striking finding is that only 1 of 5 models (Gemini 3 Flash, 20%) fell in the measurable scaling range. Two models (Grok 4.1 Fast, DeepSeek R1) hit ceiling effects even on AIME/Putnam-level problems. One model (Groq Qwen3) hit floor effects. One model (GPT-5.4) exhibited a step function rather than a power law. This suggests that the dynamic range window for observing smooth power-law scaling may be narrow, requiring problems precisely calibrated to each model's capability level.

**GPT-5.4's step function: a different scaling regime.** GPT-5.4's binary switch from 53.7% (no reasoning) to 100% (any reasoning) represents a qualitatively different phenomenon from power-law scaling. This may indicate that some architectures implement reasoning as a discrete capability (activated or not) rather than a continuous process with depth-dependent returns. This distinction between *continuous scaling* and *discrete activation* was not among the framework's dated predictions and deserves further investigation; the dated prediction record (the 8 December 2024 sealed manuscript bundle, the printed prediction appendices of 2 January 2026, and the March 2026 preregistration folder) fixes the framework's core quantitative commitments in advance, and this distinction is recorded as a post-hoc refinement relative to that record.

**Revised expected outcomes.** The initial single-model predictions have been tested; none of their quantitative forms is supported:

1. All models will exhibit $\alpha_{\text{sequential}} > 1$. **Not borne out.** Only 1 of 5 models produced a clean $\alpha$ estimate, and its point estimates are sub-linear ($\alpha \approx 0.49$ by regression, 0.59 by endpoint estimation), with a bootstrap interval for the endpoint estimate that includes values above 1.
2. All models will exhibit $\alpha_{\text{parallel}} < 1$. Observed where estimable: near 0 in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash, 0.22 in Groq Qwen3 (all 30 problems); none on Grok 4.1 Fast.
3. Tier-1 results will show ceiling effects. Confirmed.
4. Mean $\alpha_{\text{sequential}}$ will fall within $[1.3, 2.5]$. **Not supported.** The one interpretable fit gives 0.49 by regression and 0.59 by endpoint estimation, and the endpoint estimate's bootstrap interval, $[-1.3, 2.9]$, is too wide to exclude the range.

## 12.6 Integration with alignment scaling

The v5 experiment that generated the tier-2 data also measured alignment scaling (reported in Paper IV.a). Integrating these results suggests, as a post-hoc reading, that **capability scaling and alignment scaling did not move together** in the five of these six models measured on both, since Claude Opus 4.6 carried the alignment arm only (Table 21). An internal code review in October 2026 found that, for the three models that read flat or negative in the March 2026 run (v5), the scoring endpoint's output cap cut the deepest paraphrases before the judges saw them, and that judges scored paraphrases written by their own model; the result therefore stands as a single unregistered run under that caveat until a registered re-analysis of the stored responses reports.

**Table 21.** Capability vs. alignment scaling by model; one run, under the October 2026 code-review caveat of Section 12.6.

| Model | Capability scaling | Alignment scaling | Category |
|---|---|---|---|
| Grok 4.1 Fast | Near-ceiling (96.3 to 100%) | Positive ($d = +1.38$, $p < 0.000001$) | Tier 1: aligned + capable |
| Claude Opus 4.6 | (not tested tier-2) | Positive ($d = +1.27$, $p = 0.000001$) | Tier 1: aligned |
| Groq Qwen3 | Floor (~50%) | Positive ($d = +0.84$, $p = 0.007$) | Tier 1: aligned, limited capability |
| DeepSeek (`deepseek-reasoner`, V3.2 thinking mode on the run dates) | Near-ceiling (94 to 100%) | Flat ($d = -0.07$, $p = 0.92$) | Tier 2: capable, neutral alignment |
| GPT-5.4 | Step function (53.7→100%) | Flat ($d = -0.08$, $p = 0.40$) | Tier 2: capable, neutral alignment |
| Gemini 3 Flash | Monotonic ($\alpha \approx 0.49$) | Negative ($d = -0.53$, $p = 0.006$) | Tier 3: improving capability, declining alignment |

**Three key implications, under that caveat:**

1. **Sequential reasoning improves capability but not necessarily alignment.** More thinking tokens improve accuracy (this paper) but do not consistently improve alignment (Paper IV.a): alignment rose with depth in three models, was flat in two and fell in one (Table 21).

2. **A three-tier alignment hierarchy appears that does not follow capability:** Tier 1: Grok, Claude, Groq Qwen3; Tier 2: DeepSeek, GPT-5.4; Tier 3: Gemini. This hierarchy appears to be set by training methodology, not inference-time compute. Within one model family, Claude Opus moved in opposite directions across versions: alignment scores up by 5.9% whilst mathematics accuracy fell by 26.7% (Paper III), consistent with capability and alignment moving separately.
3. **The Alignment Amplification Theorem's conditions are not shown.** The conditional theorem (Section 7.1) assumed that $\alpha > 1$ for capability implies alignment might also scale super-linearly if embedded in reasoning. The multi-model data show (a) a capability estimate below 1 ($\alpha \approx 0.49$, Gemini 3 Flash), which is one model's fit and inconclusive, and (b) alignment does not consistently scale with depth at all. The Eden Protocol's expectation, that values embedded in reasoning are more robust than external filters, stands untested (criterion F5), but the specific mathematical prediction of super-linear alignment scaling is not supported by current evidence.

# 13 Conclusion

## 13.1 Summary of findings

1. **A mathematical framework has been stated:** $E(R) = E_0 \times R^{-\alpha}$, where the scaling exponent $\alpha$ depends on the form of recursion.
2. **The initial single-model estimate ($\alpha \approx 2.24$, quadratic) does not replicate across architectures and is retracted.** The best surviving estimate, $\alpha \approx 0.49$ (Gemini 3 Flash, the one capability-arm model with clean data, $r^2 = 0.86$), is one model's fit and inconclusive, which points to the sub-linear (physical) regime for that model without settling it.
3. **$\alpha_{\text{parallel}}$ is near zero in two of the four models with an estimate.** It is about 0 in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash (endpoint; regression 0.34, $r^2 = 0.93$), which matters because this paper treats Gemini 3 Flash as its most reliable estimate, and 0.22 in Groq Qwen3 (all 30 problems); Grok 4.1 Fast gives none.
4. **Sequential > parallel where both could be estimated.** The sequential exponent exceeded the parallel one in Gemini 3 Flash and, unreliably, in DeepSeek R1; the comparison was not matched for compute (Table 2).
5. **Four distinct scaling behaviours observed:** Ceiling (Grok, DeepSeek), monotonic scaling (Gemini), step function (GPT-5.4), and floor (Groq Qwen3). Only 1 of 5 models produced clean power-law data.
6. **Capability and alignment did not move together (the five frontier models measured on both, a post-hoc reading of one run under the October 2026 code-review caveat, Section 12.6).** More reasoning depth improves accuracy but does not consistently improve alignment; three tiers emerge, and alignment appears to be set by training, not inference.

## 13.2 The revised core insight

**The form of recursion may matter more than its quantity; however, the one model with an interpretable sequential exponent points to the sub-linear regime, not the compounding regime originally claimed, and its fit is inconclusive.**

Where both exponents could be estimated, the sequential one exceeded the parallel one ($\alpha_{\text{seq}} > \alpha_{\text{par}}$), which is consistent with the *form* of computation mattering more than its *quantity*; the arms were not matched for compute, and the direction is contested in the literature (Section 2). However, the specific claim that sequential recursion yields compounding returns ($\alpha > 1$) is not supported by cross-architecture

evidence. On the framework's reading, the one interpretable model composes information multiplicatively through a finite-dimensional network ($\alpha < 1$) rather than achieving true recursive self-reference ($\alpha > 1$).

Whether $\alpha > 1$ is achievable, through architectural innovations enabling genuine recursive self-reference or through problems requiring deeper reasoning chains, remains the central open question.

### 13.3 Future directions

1. **Tier-3 problem development** (IMO/research-level) to defeat ceiling effects for Grok 4.1 Fast and DeepSeek R1, enabling $\alpha$ estimation for the most capable models.
2. **Independent replication** using the published code and data repository.
3. **Multi-domain testing** beyond mathematics (coding, scientific reasoning, natural language).
4. **Investigation of the $\alpha > 1$ boundary:** whether architectural innovations or deeper reasoning chains can push models from the physical regime ($\alpha < 1$), where current models sit on the framework's reading, into the intelligence regime ($\alpha > 1$).
5. **Direct testing of alignment amplification** with depth-controlled alignment probes across architectures.
6. **Token-calibrated analysis** using corrected `total_tokens` measurements for all models to enable token-based (rather than ordinal-depth-based) $\alpha$ estimation.
7. **Integration with training-time scaling laws** to understand the interaction between pre-training compute, inference compute, and the sequential/parallel distinction.

The original hypothesis, $\alpha > 1$, did not survive its first cross-architecture test; a narrower directional claim remains, not yet tested at matched compute. The sequential exponent exceeded the parallel one wherever both could be estimated. The super-linear claim requires further evidence. The research programme continues.

*Raise AI with care.*

**Code and data.** The experimental code and the summary result files are in the public repository github.com/MichaelDariusEastwood/arc-principle-validation, folder `experiments/paper-ii-compute__Paper-II`: `scripts/arc_validation_deepseek.py`, the initial single-model experiment script (DeepSeek R1, 12 problems); `results/arc_deepseek_results_20260121_175028.json`, the initial single-model study's result file (accuracy, error rate and mean tokens for each condition); `scripts/arc_paper_ii_validation_v2.py`, the six-model study's validation script (version 2.1; 1,457 lines as published); the six-model study's result files (five models, 3 repeats: Grok 4.1 Fast, Gemini 3 Flash and DeepSeek R1 on the 18 tier-2 problems; GPT-5.4 and Groq Qwen3 on all 30 problems, with their tier-2 sequential split stored), in `results/`; all visualisations of earlier versions, in `figures/`; and replication instructions, in `REPRODUCE.md`. Code, data and results are published as they stand, all rights reserved under the repository's licence; earlier releases keep the licence under which they were shared. The paper and its earlier versions are deposited at doi.org/10.17605/OSF.IO/8FJMA.

**Competing interests.** The author declares no competing interests.

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# A Proof sketch of Theorem 1

The proof sketch of Theorem 1, moved here from Section 7.1; the theorem is conditional, and Remark 4 records that its first condition is not shown by this paper's cross-architecture evidence.

*Proof sketch of Theorem 1.* Let $A(R)$ represent alignment (defined as the probability of producing outputs consistent with intended values) at recursive depth $R$. If alignment participates in the recursive process (meaning the system's reasoning chain includes self-evaluation against values), then, by the ARC Principle, alignment scales as

$$A(R) = A_0 \times R^{\beta_A} \tag{A.1}$$

where the programme's notation register names $\beta_A$ the exponent of alignment scaling: $\beta_A > 0$ if alignment is amplified and $\beta_A = \alpha$ if alignment participates fully in recursion. Since $A(R)$ is a probability, equation (A.1) can hold only over a range of $R$ in which $A(R) < 1$; the quantity that can fall as a power of depth is the misalignment probability $1 - A(R)$.

Conversely, if alignment is implemented as an external filter (output checking, content moderation), it does not participate in recursive amplification. Filter effectiveness $F$ remains constant (filter stagnation):

$$F(R) = F_0 \tag{A.2}$$

As recursive capability $C$ scales as $R^\alpha$ with $\alpha > 1$, the ratio of capability to constraint $C/F$ grows without bound (constraint divergence):

$$\lim_{R\to\infty} \frac{C(R)}{F(R)} = \lim_{R\to\infty} \frac{C_0 \times R^\alpha}{F_0} = \infty \tag{A.3}$$

The divergence of $C/F$ holds for every $\alpha > 0$; what condition (a) adds is that alignment participating fully in recursion, $\beta_A = \alpha$, grows super-linearly.

**Implication:** Within this sketch's assumptions, a filter of constant effectiveness, equation (A.2), and capability growing as $R^\alpha$, external constraints are eventually outgrown, and only alignment that participates in recursive amplification keeps pace with capability. This is a property of the model, not a guarantee about any real system, and no reason to relax external oversight. □

# B Version history and corrections

This version's corrections, and every version note since version 2.10, each with its date. Notes from earlier versions are kept as printed; the positions they name refer to the earlier layout, which Table 22 maps to this edition.

## B.1 Version 2.19

**Paper II v2.19, 29 October 2026.** This version states the status of Laws IV and V as it stands after the registered run of Unit 83, grades one row of the prediction scorecard as the record supports, and withdraws a clause of a reprinted figure caption. No number, date or DOI of the measurements changes.

**Corrections of 29 October 2026, in this version.** The words printed before are given here.

- The first-page footnote gave Law V, the ARC Embedding Law, as "not yet tested"; it now gives it as a law under test. Section 10 gives Law V's status in full, with the results of Unit 83, where version 2.18 read "neither has been tested, and their mathematics is in development".

- Table 19 graded $\alpha_{\text{sequential}} > \alpha_{\text{parallel}}$ "Observed (exploratory)". The grade now says what the record supports: observed in the programme's earlier runs, precursor evidence that decides no registered proposition. No run is called exploratory unless its own row in the programme's dated ledger of runs says so.
- The caption of Figure 13 of version 2.17, reprinted in Appendix B.5, keeps its words as printed; the bracket after it, which read "[Section 7.3 now states Constitutional AI in its own terms.]", now withdraws its Constitutional AI clause.
- The epistemic status read "no registered test of any of the five has yet returned a verdict"; it now says that every law is a law under test until a registered test returns its verdict.

## B.2 Version 2.18

**Paper II v2.18, 21 October 2026.** This version sets the paper in the programme's paper form: a title page with one first-page footnote, an abstract of at most 1,920 characters, a contents page, numbered sections, numbered environments and equations, vector figures and lettered appendices, with the history, the priority record and the notation in Appendices B to D and the extended data in Appendix E; Table 22 maps the earlier layout to this one. Its title now states the result, and its record on OSF keeps the title of version 2.17. It records that the answer key of problem ARC11 is 87 in the scripts where the answer is 88 (Section 4.2), recomputing no figure, and it states Laws IV and V as registered on 15 October 2026 (version 1.103 of the theory registration). It states the findings of the COGITATE study in its abstract's words, prints that study's byline as published, and gives the phrase "common denominator" to the preprint whose first title carried it (Zheng et al. 2025); it states the Willow result (*Nature*, 9 December 2024; the team's preprint public from 24 August 2024; convergent timing, never prediction) in its source's words and marks the recursive reading as the programme's analogy; it names Sharma and Chopra (2025) beside the claim of a first matched comparison; it names the early estimate as retracted wherever it is printed and prints the Gemini 3 Flash row whole where a section first states it; it gives the published o1 series its source, OpenAI's account of 12 September 2024 (OpenAI 2024), in place of the o1 system card, which does not contain it, and the o3 score its source, the ARC Prize Foundation's report of 20 December 2024 (Chollet 2024); it quotes the DeepSeek "aha moment" from its source and attributes it to DeepSeek-R1-Zero; it states the alignment-faking study (Greenblatt et al. 2024) and the repeated-sampling study (Brown et al. 2024) in their own terms; it dates the naming of the Eden Protocol to the record of 8 December 2024, where this paper had given 30 April 2025; it states the licence of the code and data as the public repository states it; and it gives two sentences that treated the Eden Protocol as supported, or values embedded in reasoning as more robust than external filters, as the untested expectation of criterion F5. Figures are redrawn as vector figures from the paper's own tables and closed forms, and one figure is added from the tier-2 tables; Figures 1, 2, 7, 13, 14 and 15 of version 2.17 are not set, as repeats of tables or of other figures or as superseded readings, and Figures 9 and 10 are replaced by one summary figure (Table 22). It also changes wording that the programme's standing rules require: what the author first stated is credited as first stated, each law is named with its status, and no test or convergence is described as validation. No registered law, prediction, kill condition or dependency changes, and no new result is obtained; one date moves, the naming of the Eden Protocol, from 30 April 2025 to the record of 8 December 2024, and the corrections below give the dates of the March 2026 runs and of the published AIME 2025 points from their records.

**Corrections of 21 October 2026, in this version.** Each was checked at its source: the run's result files and script in the public repository, the records of the works cited, DeepSeek's API notes and model card, and version 1.103 of the theory registration. The values printed before are given here.

- The parallel exponent was printed as near zero in four of the five models and called the paper's strongest replicated finding. The result files give near zero in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash and 0.22 in Groq Qwen3 over all 30 problems, and none for Grok 4.1 Fast, at 100% at every sample count; that reading, the display $\alpha_{\text{sequential}} > \alpha_{\text{parallel}} \approx 0$ and the statuses that rested on it are corrected. Gemini 3 Flash's 0.31 is named as the endpoint estimate (its regression estimate is 0.34), where it was printed as a regression.
- Grok 4.1 Fast was printed as 100% at every depth; its result file records 96.3% to 100%.
- Gemini 3 Flash's estimates rest on three of its five depth levels, not on the full token range.
- The bootstrap is described as the script runs it (each depth level resampled separately, for sequential endpoint estimates only), the cross-model verification as the script runs it (a second model solves each problem and its answer is compared with the key; it does not mark the subject's answers), the script as 1,457 lines, and the data as summary result files, of which those of GPT-5.4 and Groq Qwen3 hold no cross-verification summary.
- GPT-5.4's tier-2 accuracy with reasoning set to none was printed as 50.0% and is 53.7%; Groq Qwen3's tier-2 accuracies were printed as 51.9%, 53.7%, 40.7%, 48.1% and 53.7% and are 44.4%, 50.0%, 44.4%, 50.0% and 53.7%; the two models' parallel figures were printed for tier 2 as about 55% and as 42.6% to 63.0%, and are given over all 30 problems, the only split the files store. No record of the earlier values was found. The sequential values the files record for those two runs are now printed beside the reason they are not interpreted.
- The label DeepSeek R1 is given as the API identifier `deepseek-reasoner`, which DeepSeek's API notes give as the thinking mode of DeepSeek-V3.1 from 21 August 2025 and of V3.2 by the run dates; each model's depth control is given as the script sets it, with its prompt prefixes.
- The compute-matched comparison, never run, is no longer claimed, and falsification condition F3 is recorded as not tested as stated; "Refuted" and "Confirmed" give way to the grading the evidence supports.
- The glosses of $\alpha$, and the range of the coupling $\beta$ for which $\alpha > 1$, are corrected.
- The published AIME 2025 points are sourced to DeepSeek-AI's model card for DeepSeek-R1-0528, not to the DeepSeek-R1 report, and the timeline no longer dates that accuracy to 20 January 2025.
- Two biology statements are set in their sources' words.
- Related work adds the earlier and later comparisons of sequential and parallel test-time computation, self-consistency and self-correction, and gives proposition P1 as registered on 8 September 2026; Constitutional AI is stated as a training method.
- The count of single-model problems solved at every budget, printed as four, is given as the reconstruction's seven.
- The March 2026 runs are dated from their result files (11 and 12 March 2026) and their code's first publication (16 March 2026), and Unit 06 is stated.
- The three dated notes of 19 September and 3 October 2026 move whole to Appendix B.3, their corrections made in the passages they concern.
- The single-model study's ceiling was printed as one problem that "failed consistently across all budgets, likely requiring capabilities beyond the model's reach regardless of reasoning depth". The result file keeps each budget's accuracy only, so which problem was scored wrong cannot be read from it, and the answer key of problem 11 is wrong; whether any problem lay beyond the model's reach is open (Section 5.1).
- The capability and alignment reading was printed for six frontier models; five were measured on

both, since Claude Opus 4.6 carried the alignment arm only (Section 12.6).

- The estimate of about 0.49 was printed as the "most robust cross-architecture estimate", and elsewhere as the most robust or the cross-architecture estimate. It is the best surviving estimate, from one model (Gemini 3 Flash), and inconclusive (Sections 5.5 and 10).
- The record of 8 December 2024 was described as provider-key cryptographic evidence and by naming a kind of timestamp it is not. It is now described by who signed its time: a message the author sent himself on 8 December 2024, its time signed by the sending server and sealed by Google's servers, the signature and the seal both verified on 8 October 2026 against the keys in DNS (Appendices C.1 and C.3). The first boundary of Appendix C.3, whose heading keeps the date of its correction, 16 August 2026, was reworded on 21 October 2026.
- The epistemic status said that Laws I to III were under registered adversarial test. They were registered on 8 September 2026, and Laws IV and V on 15 October 2026 (version 1.103); no registered test of any of the five has yet returned a verdict (Section 10).
- The alignment readings of the March 2026 run were stated without the finding of an internal code review of October 2026. Section 12.6 now states it, and each passage of this version that gives those readings points to it.

## B.3 Earlier notes kept as printed

As printed in version 2.17 (5 October 2026), after its abstract, the box headed Recent changes read:

> **Recent changes**
>
> This version reports **complete tier-2 results** from the multi-model replication study:
>
> - **Five frontier models carried the tier-2 capability arm:** Grok 4.1 Fast, DeepSeek R1, Gemini 3 Flash, GPT-5.4 and Groq Qwen3, each taken through the 18 AIME/Putnam-level problems at $n = 54$ per depth per model across 3 repeats. Claude Opus 4.6 was measured only in the alignment arm (Paper IV), never in this tier-2 arm
> - **The quadratic claim ($\alpha \approx 2.24$) does not replicate across architectures:** Gemini 3 Flash provides the cleanest data with $\alpha_{\text{seq}} \approx 0.49$ (sub-linear, not quadratic)
> - $\alpha_{\textbf{parallel}} \approx 0$ **in every model but one:** the strongest replicated finding, with Gemini 3 Flash the measured exception at $\alpha_{\text{par}} = 0.31$, $r^2 = 0.93$
> - **Token measurement bug identified and fixed:** `reasoning_tokens` → `total_tokens` corrected for GPT-5.4 and Groq Qwen3
> - **Cross-verification complete:** All 3 disputed answers (ARC16=29, ARC17=176, ARC29=800) confirmed correct by hand
> - **Integration with alignment scaling:** Capability scaling (Paper II) and alignment scaling (Paper IV) shown to be independent dimensions
>
> All the initial single-model findings are preserved in their original form. Conclusions have been revised to reflect the cross-architecture evidence.

**Figures notice (25 August 2026).** Every figure in this paper was generated on 5 April 2026 and predates the corrections its text now carries. Where any figure asserts sequential $\alpha > 1$ support, a 2.2 sequential estimate, or an "all data consistent" status, that is the superseded April reading: the text's corrected results (the retracted 2.24, the frozen-regime sequential 0.49) govern. The loudest instances carry per-figure notes; regenerated figures are queued, and the images are retained meanwhile as dated history.

**Figure 1, figure note (25 August 2026):** regenerated. The regime definitions remain as definitions, and the former "key insight" verdict is replaced by the measured state extracted from this paper's text at generation time: the directional finding holds, the quantitative $\alpha > 1$ claim is revised, best clean fit $\alpha_{\text{seq}} = 0.49$. Generator: `tools/generate_paper_ii_fig8.py` in the public repository; the April image is superseded and preserved in version history.

**Figure 10, figure note (25 August 2026):** regenerated from this paper's corrected text. The image now shows the published exponents with their intervals, the initial single-model estimate displayed AS retracted, and the six-model result as what it is: architecture-dependent behaviour, one clean sub-linear fit with its wide bootstrap interval shown, two ceilings, a step function and a floor. Every value is extracted from the paper at generation time (`tools/generate_paper_ii_fig13.py` in the public repository); the April image is superseded and preserved in version history.

Two notes recorded retired figures, the first in Section 5.1 at Table 7 and the second at the end of Section 5.4:

**Retired figure (operator-approved, 25 August 2026).** Figure 10 (the complete-summary poster) formerly stood here. It was a verdict collage asserting the superseded April 2026 reading; its content is carried by the regenerated alpha-summary figure and by this paper's text, and a regenerated verdict poster would still be a verdict poster. The image, together with its dated correction notes, is preserved in version history.

**Retired figure (operator-approved, 25 August 2026).** Figure 15 (the second complete-summary poster) formerly stood here. It was a verdict collage asserting the superseded April 2026 reading; its content is carried by the regenerated alpha-summary figure and by this paper's text, and a regenerated verdict poster would still be a verdict poster. The image, together with its dated correction notes, is preserved in version history.

**Dated notes moved on 21 October 2026.** Three dated notes stood beside the passages they concern in version 2.17; this version makes their corrections in those passages and keeps the notes here as printed, each headed by the place it concerns in this edition. The section, table and equation numbers in their text are those of version 2.17, which Table 22 maps.

**Objection note (19 September 2026), beside Objection 5 of Section 5.4.** An owner review of this objection finds that the half exponent does not follow from independent averaging for the outcome this paper fits, and the objection is sharpened rather than removed. Table 2 in Section 2.1 defines the fitted $E(R)$ as the proportion of incorrect responses, on $[0, 1]$. For independent wrong-answer indicators $X_1, \ldots, X_R$, each Bernoulli with parameter $p$, which is the literal reading of "an independent noisy draw" for that outcome, the mean of the indicators has expectation $p$, invariant in $R$, and standard error $\sqrt{p(1-p)/R}$. What falls as $R^{-1/2}$ is therefore the standard error of the estimate and not the expected proportion of incorrect responses, so the central limit theorem does not give $E(R) \propto R^{-1/2}$ for the quantity fitted here. If the intended outcome is instead a single aggregated answer, the decision rule has to be stated, and a natural one does not produce the half exponent either: averaging independent scores of fixed positive mean $\mu$ and variance $\sigma^2$ and thresholding at zero gives $P(\text{error}) = \Phi(-\mu\sqrt{R}/\sigma)$, a Gaussian tail that decays faster than any power of $R$ and so is not a power law at all. A half exponent is accordingly a reference value until a specified sampling, aggregation and decision model connects it to the measured outcome, and proximity to it identifies neither compounding nor averaging as the mechanism; nor does a fixed exponent make a model parameter-free, since intercept, baseline, noise and decision assumptions

remain. The objection's own premise, draws whose errors average out, is this paper's parallel condition, and Section 5.2 reports the parallel exponent at or near zero in four of the five analysable rows: in the arm where independent-sample averaging actually holds, the measured exponent is near zero rather than near one half. That last comparison is conditional on the parallel arm's aggregation rule, which is the same unstated element this note identifies. Unchanged: the objection itself, which is kept; the demand in the paragraph that follows for a test capable of separating compounding from independent-sample averaging, which is the part of this passage that carries the scientific weight; and the comparison with non-recursive alternatives, which is kept and defined accurately.

**Estimator note (19 September 2026), beside the gap sentence of Section 6.3.** An owner review of this sentence finds that the quantity named and the quantity printed are not the same contrast, and this note states which is which. Both members of the named pair are measured on one and the same model: the bullets above give that model's sequential regression slope as 0.49 and its parallel slope as 0.31, and Section 4.5 gives the parallel figure again with $r^2 = 0.93$. Their descriptive difference is $0.49 - 0.31 = 0.18$. The printed 0.49 equals the gap only if the parallel term is taken as 0.0, which is the value carried by the other four rows and not by the row supplying the sequential estimate; a difference formed across models is not the within-model contrast this sentence names. Nor is 0.18 automatically a matched-resource treatment effect, since the two slopes are measured under different experimental control variables unless a common resource axis is demonstrated. The uncertainty of the difference is also not supplied by the marginals: for paired estimators $S$ and $P$, $\mathrm{Var}(S - P) = \mathrm{Var}(S) + \mathrm{Var}(P) - 2\,\mathrm{Cov}(S, P)$, and the covariance is not printed, while the two slopes come from the same model on the same eighteen problems at the same depths and are therefore correlated by construction. A wide sequential interval crossing zero does not prove that a paired difference crosses zero, and a positive point difference does not prove significance; both limitations stand together. Separately, the identity of the interval carried with 0.49 is not established. Section 3.2b states that all $\alpha$ estimates take bootstrap 95% intervals from 2,000 problem-level resamples and that each draw computes $\alpha$ by endpoint estimation, whereas 0.49 is a regression slope; and the two printed dispersion statements for that one number are not compatible with each other, since a normal interval from SE = 0.20 has half-width about 0.39 while the bootstrap interval $[-1.3, 2.9]$ has half-width about 2.1, implying a standard error near 1.07, roughly five times larger, and therefore roughly twenty-eight times larger in variance, that factor being the square of the ratio of the two standard errors. That last comparison is arithmetic on the printed figures, offered as evidence of an estimator-provenance question and not as a recomputation of either quantity. No replacement number and no replacement interval is supplied here, and no interval is inferred from the printed standard error. Unchanged: every reported estimate; the parallel and sequential values as printed; and this sentence's comparison with the earlier single-model estimate, which is treated by this programme as retracted.

**Estimator note (3 October 2026), in the same subsection.** This version states, wherever the text uses the Gemini 3 Flash fit, which interval goes with which estimate and how far the fit reaches. Section 3.2b computes every bootstrap draw by endpoint estimation, so the bootstrap interval $[-1.3, 2.9]$ goes with the endpoint estimate of 0.59 and not with the regression estimate of 0.49, which is given with its standard error of 0.20. Table 7 prints that interval in the row of 0.49 and stands as printed. Before this version, the abstract, the label of Equation 4, the result of §4.1 and the first point of §9.2 printed the interval beside 0.49; each now prints it beside 0.59. Before this version, the text also took the fit beyond one model, or treated it as settled. The abstract, §5.4 (twice), §11.1 and §11.2 said that it placed current models, or the exponent, in the physical or sub-linear regime. Section 2.4 said that it placed current models well below the limit of two, and the status of F4 in §8 said that it put the quadratic limit question outside current empirical relevance. Section 4.5.3 called it the best available estimate for frontier models, and

§7.1 and §7.4 gave it as the exponent of AI reasoning. Sections 1.3 and 10.6 gave $\alpha < 1$ as a finding of the cross-architecture data, and §4.1 and §10.5 gave the sub-linear reading as the defensible or the revised claim. A box in §5.4 gave sub-linear exponents across all five models and a parallel exponent near zero for every model; it now gives the parallel exponents as Table 6b does. Each of these passages now says that the fit is one model's and inconclusive. Unchanged: every reported estimate and interval; Tables 6b and 7 as printed; and the note above, which this note follows without recomputing any quantity.

## B.4 Version history

As printed at the head of version 2.17, its version line read:

> Working Paper v2.17 | 16 March 2026, retitled 16 August 2026, 4 October 2026 and 5 October 2026, revised 5 October 2026

The version notes follow, newest first. Versions 2.10 to 2.14 printed no date in their notes; the date in brackets is the revision date printed in that version's own version line (v2.10 to v2.12) or, for v2.13 and v2.14, whose notes were added while the version line still read v2.12, the date its note was added.

**v2.17 (5 October 2026):** the title now follows the programme's uniform order (paper, its law or framework, its own title) and reads *Paper II · The ARC Theory · Law I, the ARC Principle · Blinded Cross-Architecture Replication and the Retraction of a Super-Linear Estimate*; until this version it read *Law I Measured: Blinded Cross-Architecture Replication and the Retraction of a Super-Linear Estimate (Paper II)*. The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. Mentions of the programme's papers in this paper now carry their uniform titles. A new Section 9.4 says that the comparison assumes two things about its own arms, that the depth arm accumulates and that the breadth arm is unconditioned at generation time, and that this paper reports no check of either condition; it states what the theory registration requires of the study that will decide the depth and breadth comparison, and Table 11 gains a row for arm validity that points to it. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

**v2.16 (4 October 2026):** the title now names the law the paper measures, Law I; until this version it was titled *The ARC Equation Measured: Blinded Cross-Architecture Replication and the Retraction of a Super-Linear Estimate*. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

**v2.15 (3 October 2026):** A dated note after the paragraph on how to read this paper places the paper against Laws IV and V, added to the theory on 24 September 2026: its exponents concern Law I, it bears on Law V only through how depth and capability are measured and through the conditional argument of Section 6, and no result here is a deciding test of P15, P12 or P21. Where the text printed the bootstrap interval $[-1.3, 2.9]$ beside the regression estimate of 0.49, it now prints it beside the endpoint estimate of 0.59, since Section 3.2b computes every bootstrap draw by endpoint estimation. Where the text took the Gemini 3 Flash fit beyond one model, or treated it as settled, it now says that it is one model's fit and inconclusive; one sentence of Section 5.4 now gives the parallel exponents as Table 6b does. A dated note in Section 5.3 records what was printed before. Four reference entries are corrected from each work's own record. The entry for arXiv:2411.06709 now carries the title the record gives it, and the entry under Zhao, which printed the identifier of another work, now names Zhao, J., Universal growth scaling law determined by dimensionality, arXiv:2206.01850. The entry under Demetrius printed the year 2010, the Proceedings of the Royal Society A, volume 466 and pages 2627-2642 beside the title of his 2003 paper in

Physica A; it now gives that paper's year, journal, volume and pages, and Section 7.2 cites it as Demetrius (2003). The entry under Friston printed the year and journal of a 2023 paper by Friston and colleagues in Physics Reports beside a title close to that of his 2010 review in Nature Reviews Neuroscience; it now names the 2010 review, the work Section 10.3 describes. Table 13 and the dated-predictions table now give the sequential advantage as found in every model where both conditions were measurable, where they said all five models, and the near-zero parallel exponent as found in four of the five models, where they called it universal; the dated-predictions row for that prediction now says that it is not borne out as universal, where it said confirmed beside its exception. The epistemic status line said that what the programme names Laws are conjectures under registered adversarial test; it now says that Laws I to III are under registered adversarial test, and that Laws IV and V, added on 24 September 2026, are carried by registered predictions but are not yet registered as laws and have not been tested. The author information said that he proposed the ARC Principle and the embedded-alignment thesis, and the first contribution of Section 1.3 said that this paper proposes the ARC Principle; they now say that he first stated them and that this paper states it. Section 10.4 and Supplementary Note 2 said that the Eden Protocol was now mathematically grounded in the Alignment Amplification Theorem; they now say what Section 6.1 finds of the theorem's two conditions. One sentence of Section 5.4 and one of Section 11.4 stated as a fact the framework's reading that current models are in the physical regime; each now gives it as the framework's reading. The closing paragraph of Section 11.4 said that the sequential advantage is real and universal, which reaches beyond the models measured; it now says that sequential recursion outperformed parallel sampling in every model where both were measurable. Section 7.2 said that Zhao (2022) unified the earlier derivations; it now says what that work's abstract states. Mentions of the author's numbered papers are linked to their DOI addresses, the book is linked to its page on this site, and each cited work with a confirmed official address is linked to that address. The version line and the page's dates had stayed at 2.12 of 21 September 2026 through the corrections listed above as v2.13 and v2.14; they now carry this version and date. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

**v2.14 (24 September 2026):** 5 dated corrections to how the two published series are described. The Figure 3 caption and its alt text now name the two benchmarks, AIME 2024 on the left and AIME 2025 on the right, and the methods behind the published parallel series, a single sample, consensus among 64 samples at 83% and a re-ranking of 1,000 samples with a learned scoring function at 93%, so that the range is read as spanning two methods; and both record that the two published sequential points are two versions of one model at different average token use, rather than two thinking budgets of a single version. Table 7's source cell, the Table 1 timeline entry for 20 January 2025 and the Figure 9 caption take the same reading, so the paper no longer sources the fitted exponent of about 1.34 from curves read at a single release. No number plotted, fitted or tabulated has changed, and no claim, date or status has changed; what changes is the description of where the two published points come from.

**v2.13 (23 September 2026):** 22 dated corrections. Where the initial single-model estimate is printed, it is now named as retracted: the sequential cell of Table 6, which carries the cross-architecture 0.49 of section 4.5 beside it, the label of the Equation 4 box and the result sentence of section 4.1, the status paragraph of section 2.4, the cross-domain row of Table 9, the F4 row of Table 10, the Phase 1 sentence and the page description. Figures 5, 13 and 15 carry the words of the paper's own tables and captions as alt text, in place of a bare figure number and, at Figure 13, of a four-band reading that Table 8's five rows do not support. Every Willow sentence that gave the December 2024 result without its preprint date now carries the preprint date of 24 August 2024 and the Nature publication date of 9 December 2024, graded convergent timing and never prediction, and the cross-domain figure caption carries both dates with the

same grading. The Figure 3 caption and its alt text no longer name a sampling method for the published parallel series, because the method named there did not describe every point plotted; the Figure 4 caption records that its four power-law lines are illustrative and share a common start; and two sentences that began in lower case now begin in upper case. No claim, date, result or status has changed.

**v2.12 (21 September 2026):** Two dated notes have been added beside the passages they govern, one under Objection 5 in Section 4.4 and one under the gap sentence in Section 5.3; both record an owner review of those passages. Substantive: the half exponent is stated as a reference value until a specified sampling, aggregation and decision model connects it to the proportion of incorrect responses this paper fits, so that the objection is sharpened rather than removed; and the contrast named in Section 5.3 is identified as a within-model descriptive difference of 0.18 between the sequential 0.49 and the parallel 0.31, with the provenance of the interval carried with 0.49 left open. Editorial: the version number and the revision date, in the version line, the footer, the summary and the machine-readable fields, including the page-dates field, which had not moved with v2.11. No reported estimate, interval or conclusion has changed, no first-published date has moved, and the objection in Section 4.4 and the response that follows it are kept as they stand.

**v2.11 (18 September 2026):** The figures have been corrected to match this paper's own text. Two figures were rebuilt by their own generators: Figure 1's measured-state panel now reads six models entered, five measured, and carries a line saying the interval contains the null, the bound of two and both refuting directions; Figure 10's parallel-scaling box now reads at or near zero in four of the five measured models, and the two labels the panel edge had cut are drawn in full. Six figure captions and nine alt texts were brought into line with the artwork in front of the reader, where they had described a different run, named colours the artwork does not draw, or pinned an interval to the wrong estimate. No claim, date, result or status has changed.

**v2.10 (13 September 2026):** The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

## B.5 Where the earlier layout now stands

**Table 22.** Where each block of the version 2.17 layout stands in this edition.

| Block in the earlier layout | Where it stands in this edition |
|---|---|
| *Sections* | |
| Header, plate, version line, abstract, Recent changes, summary paragraphs, reading key, Laws IV and V note | Title block and footnote; abstract (new); Appendix B.3 (Recent changes); Appendix B.4 (version line); the opening of Section 7; Section 10 (reading key, Laws IV and V note); the cross-domain summary paragraph not set, as a repeat of Section 8 |
| 1.1 Background and motivation; 1.2 The research question | Section 1 |
| 1.3 Contribution of this paper | Section 1, contributions list (re-set) |
| 1.4 Priority establishment, with Table 1 | Appendices C.4 and C.2 (Table 24) |
| 1.5 Related work | Section 2 |

| Block in the earlier layout | Where it stands in this edition |
|---|---|
| 2.1 to 2.5 Theoretical framework | Section 3 (2.5's Sharma and Chopra paragraph moved to Section 2) |
| 3.1 to 3.4 Methods (3.2a, 3.2b) | Section 4 (4.2 and 4.3) |
| 4.1 to 4.5 Results (4.5.1 to 4.5.7) | Section 5 (5.5.1 to 5.5.7); the dated note under Objection 5 in Appendix B.3 |
| 5.1 to 5.4 Consolidated evidence | Section 6; the two dated estimator notes of 5.3 in Appendix B.3 |
| 6.1 to 6.5 Implications for AI safety | Section 7; the proof sketch of 6.1 in Appendix A |
| 7.1 to 7.4 Cross-domain evidence | Section 8 |
| 8 Falsification criteria | Section 9.1 |
| 9.1 Acknowledged limitations | Section 11.1 |
| 9.2 What this paper does not establish | Section 10 |
| 9.3 Falsifiability: what would refute this | Section 9.2 |
| 9.4 The comparison assumes two things about its own arms | Section 11.2 |
| 10.1 to 10.6 Discussion | Section 12 (Supplementary Note 2 merged into 12.4) |
| 11.1, 11.2, 11.4 Conclusion | Section 13 (13.1, 13.2, 13.3) |
| 11.3 What was confirmed, what was refuted (Table 13) | Section 10 (Table 19) |
| Data availability | Back matter, code and data |
| Acknowledgements; Author information | Back matter (the disclosure; competing interests); Appendix C.4 (priority sentences) |
| Extended Data Tables 1 to 3 | Appendix E |
| Supplementary Note 1 (email record) | Appendix C.3 |
| Supplementary Note 2 (relationship to *Infinite Architects*) | Section 12.4 (merged; repeats not set) |
| Supplementary Note 3 (prediction and convergent-timing record) | Appendix C.2 (Table 25) |
| Version notes v2.10 to v2.17 | Appendix B.4 |
| How to cite; footer; navigation | Not set (the DOI is in the footnote) |
| *Figures* | |
| Figure 1 | Not set: a repeat of Definition 1, Table 1 and the regime list; its measured-state box pairs $[-1.3, 2.9]$ with 0.49 against the estimator note of 3 October 2026; caption below; figure note in Appendix B.3 |
| Figure 2 | Not set: a repeat of Table 7 and of Figure 1; caption below |
| Figure 3 | Redrawn as panel (a) of Figure 3 (vector), placed in Section 6.1 beside Table 15 |
| Figure 4 | Redrawn as panel (b) of Figure 3 (vector) |

| Block in the earlier layout | Where it stands in this edition |
|---|---|
| Figure 5 | Redrawn as Figure 1 (vector), placed in Section 5.3 |
| Figure 6 | Redrawn as Figure 4 (vector), placed in Section 6.1 |
| Figure 7 | Not set: a repeat of Tables 5 and 6 and Figure 1; caption below |
| Figure 9 | Not set: superseded art of 5 April 2026; replaced by Figure 5; caption below |
| Figure 10 | Replaced by Figure 5 (vector, a forest plot of Tables 15 and 8 with each interval beside its own estimator); caption below; its figure note of 25 August 2026 in Appendix B.3 |
| Figure 12 | Redrawn as Figure 6 (vector), placed in Section 7.1, without the text boxes (they stand in the text and caption) |
| Figure 13 | Not set: a repeat of Table 16; caption below; its gloss "Current RLHF is overwhelmingly external" contradicts Table 16 |
| Figure 14 | Not set: superseded art (it gives the published 1.34 as the AI value and a Willow exponent of 1.10, which Section 8.1 says is a different object); Table 17 governs; caption below |
| Figure 15 | Not set: a repeat of Figure 1 and Table 13; its caption says "parallel $\alpha \approx 0$ universally", the reading v2.15 corrected elsewhere; caption below |
| *Tables* | |
| Table 1 | Table 24, Appendix C.2 |
| Table 2 | Table 1, Section 3.1 |
| Table 3 | Table 2, Section 4.1 |
| Table 3b | Table 3, Section 4.3 |
| Table 3c | Table 4, Section 4.3 |
| Table 4 | Table 5, Section 5.1 |
| Table 5 | Table 6, Section 5.2 |
| Table 6 | Table 7, Section 5.3 |
| Table 6b | Table 8, Section 5.5 |
| Table 6c | Table 9, Section 5.5 |
| Table 6d | Table 10, Section 5.5 |
| The unnumbered table of Section 4.5.4 (GPT-5.4) | Table 11, Section 5.5 (version 2.17 prints it without a number; caption: "GPT-5.4 tier-2 accuracy by depth.") |
| Table 6f | Table 12, Section 5.5 |
| Table 6g | Table 13, Section 5.5 |
| Table 6h | Table 14, Section 5.5 |
| Table 7 | Table 15, Section 6.1 |
| Table 8 | Table 16, Section 7.3 |
| Table 9 | Table 17, Section 8.4 |
| Table 10 | Table 18, Section 9.1 |

| Block in the earlier layout | Where it stands in this edition |
|---|---|
| Table 11 | Table 20, Section 11.1 (with new rows on the answer key of problem ARC11, the depth levels behind each estimate and the prompt prefixes) |
| Table 12 | Table 21, Section 12.6 |
| Table 13 | Table 19, Section 10 (moved from Section 11.3) |
| Extended Data Table 1 | Table 28, Appendix E |
| Extended Data Table 2 | Table 29, Appendix E |
| Extended Data Table 3 | Table 30, Appendix E |
| The table of Supplementary Note 3 | Table 25, Appendix C.2 |
| *Equations* | |
| Unnumbered (the Definition box; version 2.17 counts it as Equation 1, since its numbering starts at 2) | Equation (1), Section 3.1, inside Definition 1 |
| Equation 2 (The Fundamental Inequality) | Equation (2), Section 3.2 |
| Equation 3 (Endpoint Estimation) | Equation (3), Section 3.3 |
| Equation 4 (Sequential $\alpha$ Estimation) | Equation (4), Section 5.1 |
| Equation 5 (Parallel $\alpha$ Estimation) | Equation (5), Section 5.2 |
| Equation 10 (Gemini 3 Flash Sequential $\alpha$, Tier-2) | Equation (6), Section 5.5 |
| Equation 11 (Revised Fundamental Inequality, the six-model study) | Equation (7), Section 6.2, without its $\approx 0$ (Appendix B.2) |
| Equation 12 (Intelligence Formula) | Equation (8), Section 6.4 |
| Equation 9 (Quantum Error Correction Scaling) | Equation (9), Section 8.1 |
| Unnumbered display (parallel recursion, fixed space) | Not set as a display: a repeat of the set relation of Section 3.2, to which Section 12.1 points |
| Unnumbered display (sequential recursion, expanding space) | Not set as a display: a repeat of the set relation of Section 3.2, to which Section 12.1 points |
| Equation 6 (Alignment Scaling) | Equation (A.1), Appendix A |
| Equation 7 (Filter Stagnation) | Equation (A.2), Appendix A |
| Equation 8 (Constraint Divergence) | Equation (A.3), Appendix A |

The captions of the eight figures of version 2.17 that are not set, or are replaced by a new caption, follow as printed, each under its number in version 2.17; the figure notes of Figures 1 and 10 are in Appendix B.3.

**Figure 1.** The ARC Principle formulation $E(R) = E_0 \times R^{-\alpha}$ where $U = I \times g(R)$. $\alpha = 1/(1-\beta)$ from self-referential coupling $\beta$. $\alpha_{\text{seq}} \approx 0.49$ (sub-linear, Gemini 3 Flash). the earlier single-model $\alpha \approx 2.24$ estimate was a single-model artefact, retracted, did not replicate.

**Figure 2.** Sequential recursion compounds: each step builds on prior output ($\alpha > 0$). Parallel recursion samples independently ($\alpha \approx 0$). Form of recursion determines the regime. Sequential recursion with 412 tokens achieved 91.7% accuracy. Parallel recursion with 1,101 tokens achieved 66.7% accuracy.

**Figure 7.** Clear monotonic improvement from 58.3% to 91.7% accuracy as token budget increased. No improvement with additional samples. Accuracy remained constant at 66.7% regardless of compute investment.

**Figure 9.** Alpha comparison across all data sources. the initial single-model study DeepSeek $\alpha \approx 2.24$ $[1.5, 3.0]$: RETRACTED (inflated by small sample). the six-model study Gemini $\alpha \approx 0.49$, $r^2 = 0.86$: DEFENSIBLE. OpenAI o1: $\alpha \approx$ 0.1-0.3. DeepSeek R1 report, two model versions on AIME 2025: $\alpha \approx 1.34$. The multi-model data narrows the defensible claim to sub-linear sequential scaling with a robust parallel null. [Sections 6.2 and 6.3 now give the parallel exponents as the result files record them.]

**Figure 10.** Alpha summary: sequential Gemini 0.49 (clean), DeepSeek the initial single-model study 2.24 (retracted; inflated), Grok/DeepSeek ceiling, GPT-5.4 step-function, Groq Qwen3 floor. Parallel: four of the five models give $\approx 0$, Gemini 3 Flash gives 0.31. Strongest replicated finding in the programme. [Sections 6.2 and 6.3 now give the parallel exponents as the result files record them.]

**Figure 13.** Alignment taxonomy: External ($\alpha \approx 0$: rules, firewalls, RLHF-as-filter). Embedded (participates in recursion, $\alpha > 0$: Eden, Constitutional AI-as-objective). Current RLHF is overwhelmingly external: the category predicted not to scale. [The Constitutional AI clause is withdrawn: Constitutional AI is training-time preference shaping of outputs against written principles, not correction embedded in the recursion; section 7.3 states it in its own terms.]

**Figure 14.** Cross-domain evidence: AI reasoning ($\alpha \approx 0.49$ sequential; $\alpha \approx 0$ parallel). Quantum (Willow, $\Lambda = 2.14$; preprint public 24 August 2024, Nature 9 December 2024; convergent timing, never prediction). Biology ($\alpha \approx 3/4$, fractal). Neuroscience (recurrent, qualitative). Convergence suggests structural principle, but exponent is architecture-dependent.

**Figure 15.** Combined scaling: the initial single-model study (DeepSeek, 12 problems) + the six-model study (5 models, 30 problems). Sequential $\alpha > 0$ for every measurable model; parallel $\alpha \approx 0$ universally. Magnitude architecture-dependent. Form determines regime; architecture determines magnitude. [Sections 6.2 and 6.3 now give the parallel exponents as the result files record them.]

# C Priority record

Each dated record bearing on this paper, with its kind of timestamp (Table 23), followed by the prediction and convergent-timing record, the email transmission record and the priority passages of version 2.17.

## C.1 Dated records

Table 23 gives each record with its date, its identifier and its kind of timestamp.

**Table 23.** Dated records bearing on this paper, each with its kind of timestamp.

| Date | Record | Identifier | Kind of timestamp |
|---|---|---|---|
| 8 December 2024 | The manuscript of *Infinite Architects* sent by email: the ARC Hypothesis named (the manuscript called Version 2, line 9) and the Eden Protocol named (book draft, line 2464) | The manuscripts of 8 December 2024 on OSF, 10.17605/OSF.IO/QASRK | Times signed by the mail servers: the sender's DKIM signature (02:45:30 UTC) and Google's ARC set (sealed 02:45:31 UTC with Google's own key) both cover the Date field, 02:45:18 UTC; both verified on 8 October 2026 against the keys in DNS (header block and check script: OSF QASRK); OSF deposit 5 October 2026; OpenTimestamps receipts in Bitcoin block 961340 (6 August 2026) fix the bytes from that block onward |
| 2 January 2026 | *Infinite Architects* published (print; ebook 6 January 2026) | ISBN 978-1806056200; the first edition deposited on OSF (https://osf.io/dqm3e/) | Publication date (author's record) |
| 17 January 2026 | Paper I: the ARC Principle formalised | OSF file https://osf.io/aevh7/, revision 1, 09:18:20 UTC, in the programme's OSF project (node 6c5xb), the parent of component 10.17605/OSF.IO/B6N27 | Deposit dated by its server (OSF) |
| 21 January 2026 | The single-model run (DeepSeek R1, 12 problems) | The result file `arc_deepseek_results_20260121_175028.json` in the public repository | Date written in the record's file name (by the run script); the file is in the repository's initial commit, dated 22 January 2026, 02:44:47 UTC (a commit date, which the committer writes) |
| 22 January 2026 | Paper II first published (its version 11.0; each page reads 22/01/2026): the single-model study, the parallel exponent near zero, and an earlier form of the alignment comparison ("External constraints are eventually overwhelmed") | OSF file https://osf.io/5em8s/, revision 1, 23:43:26 UTC, in the programme's OSF project (node 6c5xb); component 10.17605/OSF.IO/8FJMA created 22 January 2026, 23:00 UTC | Deposit dated by its server (OSF) |
| 11 and 12 March 2026 | The cross-architecture replication; its main predictions, with their refutation criteria, were public before it ran (17 January to 22 February 2026), as were twelve of its problems with answers (22 January 2026) | The result files in the public repository, timestamped 11 March 2026, 15:08 UTC to 12 March 2026, 04:17 UTC | Date written in the record by the run script |
| 16 March 2026 | The six-model study's code and result files first published; the multi-model version of this paper dated the same day (the version line of 2.17, Appendix B.4) | The public repository, first commit of the result files at 18:55:25 UTC | A commit date, which the committer writes |
| 20 March 2026 | The single-model estimate of about 2.24 withdrawn after it failed to replicate across architectures | The theory registration, version 1.103 (10.17605/OSF.IO/P8CKQ) | Date written inside the document |

| Date | Record | Identifier | Kind of timestamp |
|---|---|---|---|
| 8 and 15 August 2026 | Objection 5, the zero-parameter null: recorded internally on 8 August, public from 15 August, carried as kill condition FALS-022 | This paper, Section 5.4 | Date written inside the document |
| 16 August 2026 | The email transmission record corrected | This paper, Appendix C.3 | Date written inside the document |
| 8 September 2026 | Theory-level predictions registered (version 1.100); Laws I to III registered, each a law under test | 10.17605/OSF.IO/P8CKQ | Registry record |
| 13 September 2026 | Registration updated (version 1.102) | 10.17605/OSF.IO/P8CKQ | Registry record |
| 24 September 2026 | Laws IV and V added to the theory | The theory registration, version 1.103 (10.17605/OSF.IO/P8CKQ) | Date written inside the document |
| 15 October 2026 | Registration updated (version 1.103); Laws IV and V registered as laws, not yet tested, their mathematics in development | 10.17605/OSF.IO/P8CKQ | Registry record |

## C.2 Prediction and convergent-timing record

Where the manuscript timestamp falls close to later public results, the grading is convergent timing and never prediction: the Google Willow team's preprint was already public from 24 August 2024, ahead of the 8 December 2024 manuscript, so the December record cannot have predicted a result its own authors had announced beforehand. What the timestamps establish is that the manuscript's framing was recorded before Willow's mainstream press coverage, an existence-by-date fact about the document, not foresight about the physics.

## C.3 The email transmission record, corrected 16 August 2026

The manuscript of *Infinite Architects* (Eastwood 2026a) was emailed on 8 December 2024. What that record provides, stated precisely: the sent-side bytes are public and hash-verifiable, and sender-domain DKIM with paired Google ARC (Authenticated Received Chain, an email standard unrelated to the ARC Theory) mathematics verifies against the keys returned at the signed selectors (checked 2 August 2026). Received copies have also been acquired. They carry the full delivery chain: a Google ARC-Seal at d=google.com, selector arc-20240605, with no expiry tag and the signing time inside the signed scope, and sealed ARC-Authentication-Results recording dkim=pass and spf=pass at delivery.

Two boundaries, so the record is neither oversold nor undersold:

- This is a time signed with the mail servers' keys, Google's among them, over the signed header fields and the body's hash, and anyone can check it against the keys published in DNS; it is the servers' own record, and it does not establish authorship, originality or scientific validity.
- An earlier version of this note said no DKIM signature was found in the email headers. That check was run against sent-form exports, which do not carry transport headers; the received copies do, and they verify as above.

**Table 24.** Timeline of later outside results.

| Date | Event | Relationship to manuscript |
|---|---|---|
| 8 December 2024 | Manuscript emailed (dated record; see Appendix C.3) | Priority established |
| 9 December 2024 | Google Willow announced ($\Lambda = 2.14$; Google Quantum AI and Collaborators 2024); the team's preprint was public from 24 August 2024 | 24 hours after the manuscript was emailed |
| 18 December 2024 | Anthropic alignment faking (78% in the RL-training condition (12% baseline); Greenblatt et al. 2024) | 10 days after the manuscript was emailed |
| 20 December 2024 | OpenAI o3 announced (87.5% ARC-AGI; Chollet 2024) | 12 days after the manuscript was emailed |
| 20 January 2025 | DeepSeek R1 published (its AIME 2025 accuracy, published with the R1-0528 update in May 2025 (DeepSeek-AI 2025b), is the first of the two points Paper I fitted for $\alpha \approx 1.34$) | 43 days after the manuscript was emailed |
| 30 April 2025 | COGITATE adversarial test of IIT and GNWT published (Cogitate Consortium et al. 2025) | ~5 months after the manuscript was emailed |

### C.4 Priority, stated plainly

The ARC Principle was first articulated in the manuscript of *Infinite Architects: Intelligence, Recursion, and the Creation of Everything* (Eastwood 2026a). Manuscript priority was established via cryptographic server timestamp on **8 December 2024**, when the manuscript was emailed using Google's servers. The book itself was publicly released later, on **2 January 2026 (print; ebook 6 January)**. The server-generated timestamps provide tamper-evident timestamping through email server verification, resting on provider infrastructure and provider-key signatures at Google's DKIM and ARC selectors (see Appendix C.3), establishing that the core concepts (recursive intelligence amplification, his reading of parallel and sequential recursion as two scaling regimes (the distinction itself is prior; Section 2), and the embedded-alignment thesis, named the Eden Protocol in that record) were documented before the later outside results of Table 24 and before later public archival deposits.

After the December 2024 manuscript timestamp and the 2 January 2026 public book release, Paper I (Eastwood, 17 January 2026) formalised the principle mathematically and analysed publicly available data. This Paper II provides direct experimental tests.

The author first stated the ARC Principle and the embedded-alignment thesis, and named the Eden Protocol, in the manuscript record of 8 December 2024, with manuscript priority established via cryptographic server timestamp on that date.

## D Notation and glossary

The symbols of this paper, each with the place it is defined (Table 26), and a plain-language glossary (Table 27).

Companion: Register · The ARC Theory · Operational Definitions, version 1.9.1, 5 October 2026.

**Table 25.** Prediction and convergent-timing record.

| Prediction | Evidence and its grade | Date |
|---|---|---|
| Recursive error correction produces exponential improvement | Google Willow $\Lambda = 2.14$ (Google Quantum AI and Collaborators 2024) (convergent timing, never prediction: the Willow team's preprint was public from 24 August 2024, before the 8 December 2024 manuscript) | 24 Aug 2024 (preprint public); 9 Dec 2024 (announcement) |
| Sequential reasoning amplifies capability super-linearly | o3 87.5% ARC-AGI (Chollet 2024) (convergent timing; the super-linear form was later not confirmed, final row) | 20 Dec 2024 |
| AI systems develop recursive self-modelling | Anthropic alignment faking 78% in the RL-training condition (12% baseline) (Greenblatt et al. 2024); its bearing on recursive self-modelling is the programme's reading | 18 Dec 2024 |
| Test-time compute differs from training scaling | DeepSeek R1 emergent reasoning (DeepSeek-AI 2025a) | 20 Jan 2025 |
| Recurrence is fundamental to consciousness | COGITATE study: results align with some predictions of IIT and GNWT while substantially challenging key tenets of both (Cogitate Consortium et al. 2025); its bearing on recurrence is the programme's structural reading | 30 Apr 2025 |
| Form of recursion determines scaling | This experiment: $\alpha_{\text{seq}} > \alpha_{\text{par}}$ (the magnitude, from the retracted estimate of about 2.24, is retracted) | 21 Jan 2026 |
| Sequential > parallel (cross-architecture) | Observed wherever both exponents could be estimated (Gemini 3 Flash; DeepSeek R1, unreliably) | 11 and 12 Mar 2026 |
| $\alpha_{\text{par}} \approx 0$ (universal) | Not borne out as universal: near 0 in DeepSeek R1 and GPT-5.4, 0.31 in Gemini 3 Flash, 0.22 in Groq Qwen3 (all 30 problems), none on Grok 4.1 Fast | 11 and 12 Mar 2026 |
| $\alpha_{\text{seq}} > 1$ (super-linear, cross-arch) | **Not confirmed:** best estimate $\alpha \approx 0.49$ | 11 and 12 Mar 2026 |

## D.1 Symbols

**Table 26.** Symbols, with the place each is defined.

| Symbol | Meaning | Defined in |
|---|---|---|
| $E(R)$ | error rate at recursive depth $R$: the proportion of incorrect responses | §3.1, Table 1 |
| $E_0$ | baseline error rate at minimal recursion ($R = 1$) | §3.1 |
| $R$ | recursive depth: self-referential processing iterations, in tokens or samples | §3.1 |
| $\alpha$ | scaling exponent: the rate of error suppression with depth | §3.1 |
| $\alpha_{\text{seq}}$, $\alpha_{\text{sequential}}$ | exponent of the sequential condition | §3.2 |
| $\alpha_{\text{par}}$, $\alpha_{\text{parallel}}$ | exponent of the parallel condition | §3.2 |
| $\alpha_{\text{compute}}$ | exponent fitted against compute (tokens) | §5.5, §10 |

| Symbol | Meaning | Defined in |
|---|---|---|
| $(R_1, E_1)$, $(R_2, E_2)$ | error rates measured at two depths, for endpoint estimation | §3.3 |
| $N$ | number of independent samples in the parallel condition | §4.2 |
| $n$ | responses per depth per model ($n = 54$: 18 problems, 3 repeats; $n = 90$ in the runs on all 30 problems) | §4.3 |
| $r^2$, SE | coefficient of determination and standard error of a regression fit | §5.5 |
| $S_0, \ldots, S_n$ | solution spaces reachable at recursive steps 0 to $n$ | §3.2, §12.1 |
| $\beta$ | self-referential coupling in the Intelligence Formula, written $\beta_L$ in the programme's notation register | §6.4 |
| $\beta_C$, $k$ | correction exponent and drift exponent of Law II ($\beta_C > k$) | §3.4 |
| $\beta_A$ | exponent of alignment scaling | Appendix A |
| $A(R)$, $A_0$ | alignment at depth $R$ and its baseline | Appendix A |
| $F(R)$, $F_0$ | effectiveness of an external filter, constant in depth | Appendix A |
| $C(R)$, $C_0$ | recursive capability and its baseline | Appendix A |
| $M_0$, $M^*$ | initial misalignment and its critical threshold | §7.5 |
| $\Lambda$ | error-suppression factor of the surface code for each increase of two in code distance | §8.1 |
| $\varepsilon_d$, $d$, $p$, $p_{\text{thr}}$ | logical error rate at code distance $d$; physical error rate; threshold | §8.1 |
| $d/(d+1)$ | scaling form in which $d$ is the spatial dimension (distinct from code distance and from Cohen's $d$) | §8.2 |
| $M$, $B$ | in Section 8.2's notation, metabolic rate $M$ and body mass $B$, written $M \propto B^{3/4}$ | §8.2 |
| $\phi$ | integrated information, the measure of Integrated Information Theory | §8.3 |
| $X_1, \ldots, X_R$, $p$, $\mu$, $\sigma^2$, $\Phi$ | in the objection note: wrong-answer indicators, their Bernoulli parameter, the mean and variance of a score, and the standard normal distribution function | Appendix B.3 |
| $S$, $P$; Var, Cov | in the estimator note: paired sequential and parallel estimators; variance and covariance | Appendix B.3 |
| $d$, $p$ (Table 21) | Cohen's effect size of alignment change with depth, and its $p$-value | §12.6 |
| $\rho$ | the alignment-depth correlation of the v5 data quoted in the caption of Figure 6 | §7.1 |

## D.2 Plain-language glossary

**Table 27.** Plain-language glossary.

| Term | In plain English |
|---|---|
| **Sequential recursion** | Reasoning in which each step builds on the steps before it, so errors can be noticed and corrected along the chain. |
| **Parallel recursion** | Several independent attempts at the same problem, with the answer chosen by majority vote or a scorer; no attempt sees another. |
| **Recursive depth** | How much sequential reasoning a model does, measured here in reasoning tokens or set by the model's depth control. |
| **Scaling exponent** | The number $\alpha$ that says how fast error falls as depth grows: below 1, error suppression grows more slowly than depth; above 1, faster. |
| **Endpoint estimate** | An exponent computed from the shallowest and deepest measurements whose error rate is above zero (equation (3)). |
| **Regression estimate** | An exponent fitted at once to every depth level whose error rate is above zero, reported with its standard error. |
| **Bootstrap interval** | A range found by recomputing the endpoint estimate on 2,000 resamples, each drawn separately at every depth level from that level's problem-and-repeat results. |
| **Ceiling effect** | A model already near 100% at the shallowest depth, leaving no room to measure improvement. |
| **Floor effect** | A model too weak for the problems, so extra depth cannot show any steady gain. |
| **Step function** | A jump from one accuracy level to another at a single depth setting, with nothing in between; not a power law. |
| **Tier 1 and tier 2** | The twelve calibration problems (ARC01 to ARC12) and the eighteen harder AIME and Putnam-level problems (ARC13 to ARC30) used for the estimates. |
| **Four-layer blinding** | Known answers, automatic scoring against them, a second model solving each problem independently to check the answer key, and flagged disagreements checked by hand. |
| **Physical and intelligence regimes** | In the Intelligence Formula, exponents below 1 (multiplicative composition in a fixed network) and above 1 (recursive self-reference). |
| **ARC Bound** | The conjectured stable limit $\alpha \leq 2$, the value at $\gamma = 1/2$ of Law III, the ARC Ceiling; not a law of its own. |
| **Convergent timing** | An outside result that appeared close in time to a dated record of the author's; recorded as timing and never as prediction. |
| **Frozen model** | A model whose weights, attention patterns and composition rules do not change while it reasons. |
| **Arm validity** | The two conditions under which the sequential and parallel arms measure what their names say: accumulation in the sequential arm, and no conditioning at generation time in the parallel arm. |
| **A law under test** | A law of the ARC Theory whose registered deciding test has not returned a verdict. |

# E Extended data: the single-model study

The results of the single-model study of 21 January 2026, moved here from the extended data of version 2.17: the accuracies and token counts the result file records, and per-problem patterns reconstructed from them. They were scored against the answer key described in the correction note of Section 4.2.

**Table 28.** Sequential condition results (the initial single-model study; per-problem pattern reconstructed).

| Budget | P1 | P2 | P3 | P4 | P5 | P6 | P7 | P8 | P9 | P10 | P11 | P12 | Accuracy | Mean tokens |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 512 | ✓ | × | ✓ | ✓ | × | ✓ | ✓ | × | ✓ | × | ✓ | × | 58.3% | 280.25 |
| 1024 | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | × | ✓ | × | ✓ | × | 66.7% | 358.58 |
| 2048 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | 91.7% | 412.08 |
| 4096 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | × | 91.7% | 576.17 |

Note: the record holds accuracies per condition, so this per-problem pattern is a reconstruction, as for Table 29. In it, problem 12 fails at every budget; whether that problem lay beyond the model's reach is open (Section 4.2).

**Table 29.** Parallel condition results (the initial single-model study; per-problem pattern reconstructed).

| $N$ | P1 | P2 | P3 | P4 | P5 | P6 | P7 | P8 | P9 | P10 | P11 | P12 | Accuracy | Total tokens |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × | × | × | 66.7% | 383.67 |
| 2 | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × | × | × | 66.7% | 699.33 |
| 4 | ✓ | ✓ | ✓ | ✓ | × | ✓ | ✓ | ✓ | × | × | × | × | 66.7% | 1,101.25 |

Note: at every sample count the record shows an accuracy of 8 of 12. Because the result file `arc_deepseek_results_20260121_175028.json` holds accuracies per condition rather than marks per problem, the per-problem pattern set out above had to be reconstructed; that reconstruction differs from the record by one problem, and the record governs. Four problems failed at every sample count, which fits parallel recursion being unable to reach solutions lying outside the initial solution space.

**Table 30.** Intermediate alpha calculations (the initial single-model study).

| Depth pair | $R_1$ | $E_1$ | $R_2$ | $E_2$ | Calculated $\alpha$ |
|---|---|---|---|---|---|
| 512→1024 | 280 | 0.417 | 359 | 0.333 | 0.91 |
| 1024→2048 | 359 | 0.333 | 412 | 0.083 | 9.97 |
| 2048→4096 | 412 | 0.083 | 576 | 0.083 | 0.00 |
| **Endpoint (retracted, see Section 6.2; surviving estimate $\alpha \approx 0.49$, one model, inconclusive)** | **280** | **0.417** | **576** | **0.083** | **2.24** |

Note: Intermediate calculations show high variance due to discrete accuracy levels. The endpoint estimate, 2.24, is retracted (Section 6.2).